Ferroelectric Hybrid Processor Family

Tier-1 Engineering Specification

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This document provides the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family: a standard premium phone chip (Ph-1), a high-end desktop chip (Pc-1), and an entry-level single-socket server chip (Dc-1S). The architecture combines an RV64GCVH out-of-order CPU core, a FeFET non-volatile L3 cache, and an analog AI crossbar on a standard N5 CMOS process with a hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) back-end module. The document provides the complete engineering parameters for design, fabrication, test, and operation at the specification level: cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria (Parts 3–16), plus the register map, firmware interface, interrupt assignments, device tree and ACPI bindings, full pin description, mechanical drawings, bring-up checklist, and security provisioning procedure (Appendices A–H). The advanced optimization promotion is in Part 17. The design is released as a public, patent-free engineering specification.

This is a 6DUFT inspired technology.

Supersession Notice. This document supersedes the release of 11 September 2026. The previous release is no longer considered viable and should not be used. The new document replaces it in full. The three Tier-1 products (Ph-1, Pc-1, Dc-1S), the three die designs, the FeFET cell, the NVL3 subsystem, the analog crossbar, the CPU core, the fabrication flow, the verification gates, the operating procedures, the cost analysis, the security features, the failure modes, and the health and safety notes are all specified here. The register map, firmware interface, interrupt assignments, device tree and ACPI bindings, full pin description, mechanical drawings, bring-up checklist, and security provisioning procedure are provided as Appendices A–H.

Abstract

This document is the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family.

The architecture combines three elements:

1. **CPU:** An RV64GCVH out-of-order processor (forked from the open-source XiangShan design and extended to 10-wide and 12-wide execution) with hypervisor support, a 64-bit MMU, and a 512-bit vector unit.
2. **Non-volatile L3 (NVL3):** A FeFET-based non-volatile cache layer of 24–64 MB per die, providing instant-on wake from sleep and retained state across power cycles.
3. **AI crossbar:** An analog 2-bit-signed crossbar of 512×512 or 1024×1024 cells, providing hardware-accelerated inference for convolutional and small transformer models on the Phone and PC products.

The three Tier-1 products are:

- **Phone:** Ph-1 (4.5 GHz boost / 3.0 GHz sustained, 6 big + 4 little cores, 150 mm^2 die, 64 MB NVL3, 512^2 crossbar).
- **PC:** Pc-1 (28 cores at 6.2 GHz boost / 5.5 GHz sustained, 292 mm^2 die, 64 MB NVL3, 1024^2 crossbar).
- **Datacenter:** Dc-1S (single-die, 32 cores at 6.0 GHz boost / 5.5 GHz sustained, 205.9 mm^2 die, 24 MB NVL3).

The platform uses standard N5 CMOS with a hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition back-end module. No exotic front-end materials are required.

The document covers architecture, cell geometry and materials, NVL3 subsystem, crossbar subsystem, CPU core, fabrication flows, operating principles, performance specifications, verification and qualification, cost analysis, security features, failure modes and mitigations, health and safety notes, and the advanced optimization promotion.

The design is released openly. No patent is sought.

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Part I
Introduction and Overview

I.1 What This Document Is

This is the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family. It defines three products (Ph-1, Pc-1, Dc-1S) built from three unique die designs (Phone, PC, DC). It is the engineering reference for any party that wishes to fabricate, test, operate, or modify these products.

This document supersedes the release of 11 September 2026. The previous release is no longer considered viable and should not be used. The new document replaces it in full.

The document contains the complete engineering specification needed to design, build, verify, operate, and provision the three products: cell geometry, material composition, circuit design, fabrication sequence, verification criteria, operating procedures, thermal design, package specifications, and acceptance criteria (Parts 3–16); plus the register map for every custom block, the firmware interface protocols, the interrupt source assignments, the device tree and ACPI bindings, the full pin description, the mechanical drawings, the bring-up checklist, and the security provisioning procedure (Appendices A–H). The advanced optimization promotion is in Part 17. Companion artifacts that are outputs of the design process (RTL source, GDS, firmware source, board reference design, ATE test program) are provided separately with production engagement.

I.2 What This Document Covers

The document covers the design (Parts 2–8), the fabrication and verification flow (Parts 9–10), the operating procedures (Part 11), the performance and cost analysis (Parts 12–13), the security features (Part 14), the failure modes and health and safety notes (Parts 15–16), the advanced optimization promotion (Part 17), and the eight appendices required to build, operate, and provision the products (Appendices A–H). The full list:

- **The three Tier-1 products:** configurations, cores, clocks, power, pricing, and competitive positioning (Part 2).
- **The FeFET cell:** HZO film, silicon dioxide interface, top and bottom electrodes, geometry, threshold window, read and write operations, endurance, retention, wake-up procedure (Part 3).
- **The NVL3 subsystem:** 4096-row banks, readout circuit, reference generation, calibration, disturb budget, per-column screening (Part 4).
- **The analog crossbar:** 3T binary-weighted cell, signed weight handling, DAC/TIA/ADC array, calibration, accuracy, real-workload efficiency (Part 5).
- **The CPU core:** RV64GCVH ISA, microarchitecture, branch prediction, vector unit, MMU, hypervisor, multi-core organization, x86 translation via Box64 (Part 6).
- **The three dies:** area breakdown, per-product configurations, power budget, thermal design, packages, cooling (Part 7).
- **The three Tier-1 products in detail:** per-SKU specifications, positioning against competitors (Part 8).

- **Fabrication:** N5 front-end, FeFET MEOL insertion, mask count, thermal budget, yield model, wafer cost (Part 9).
- **Verification and qualification:** the twelve gates, acceptance criteria, test flow, reliability testing (Part 10).
- **Operating procedures:** boot sequence, power states, DVFS, thermal throttling, error handling, OS support, Windows-via-Wine compatibility (Part 11).
- **Performance:** single-thread, multi-thread, AI throughput, power efficiency, sleep power, wake time, honest comparison to Intel, AMD, Qualcomm, and Apple (Part 12).
- **Cost analysis:** wafer cost, die cost per product, package and test, per-SKU margin, cost of ownership (Part 13).
- **Security features:** secure boot chain, memory encryption, side-channel mitigations, attestation, debug authentication, tamper response, OTP allocation, firmware update (Part 14).
- **Failure modes and mitigations:** material, cell, circuit, system, chiplet, and hybrid-specific failure modes (Part 15).
- **Health and safety notes:** any real and non-obvious hazards in the production or operation of the device (Part 16).
- **Advanced optimization and higher-tier products:** the advanced optimization configurations and the higher-tier product variants available beyond the three Tier-1 products, with performance and price stated relative to the baseline (Part 17).
- **Register map:** every register in the fourteen custom MMIO blocks, with bit fields, access codes, and reset values (Appendix A).
- **Firmware interface:** the register-level protocols for NVL3 write-verify, NVL3 pre-wake-up, NVL3 calibration, crossbar weight update, crossbar calibration, firmware update, security key provisioning, and error recovery (Appendix B).
- **Interrupt source assignments:** the 64 PLIC-lite sources, their trigger types, their default priorities, and the per-product differences (Appendix C).
- **Device tree and ACPI bindings:** the full DT node structure, the ACPI table inventory, and the DSDT namespace for every custom block (Appendix D).
- **Full pin description:** the ballmap CSV format, the signal catalog, the differential pair catalog, the power/ground placement rules, and the electrical specification per interface (Appendix E).
- **Mechanical drawings:** the package outline dimensions, substrate stack-up, heatsink mounting, thermal interface specification, solder reflow profile, MSL, and board keep-out zones (Appendix F).
- **Bring-up checklist:** the 12-stage silicon bring-up procedure with per-measurement acceptance criteria and failure diagnosis (Appendix G).
- **Security provisioning procedure:** key generation, OTP injection, locking, attestation key derivation, debug unlock, decommissioning, and security event logging (Appendix H).

I.3 Reading Guide

The fabricator should read Parts 3 (FeFET cell), 4 (NVL3), 5 (crossbar), 7 (dies), 9 (fabrication), 10 (verification).

The firmware / OS engineer should read Parts 6 (CPU), 11 (operating procedures), 12 (performance), Appendix A (register map), Appendix B (firmware interface), Appendix C (interrupt assignments), Appendix D (device tree and ACPI).

The architect should read Parts 2 (products), 3–6 (subsystems), 7–8 (dies and products).

The business / program manager should read Part 2 (overview), 8 (product details), 12 (performance), 13 (cost), 14 (security), 17 (advanced optimization).

The external reader should read Part 2 (overview), 12 (performance and honest comparison), 13 (cost), 17 (advanced optimization).

The board designer should read Parts 7 (dies), Appendix E (full pin description), Appendix F (mechanical drawings).

The test / bring-up engineer should read Part 10 (verification), Appendix G (bring-up checklist), Appendix H (security provisioning).

The security engineer should read Part 14 (security features), Appendix H (security provisioning procedure), Appendix B §B.7 (security engine key provisioning).

I.4 Design Philosophy

The design uses proven components (commercial HZO FeFET non-volatile memory, the open-source XiangShan CPU core, standard analog crossbar cells), open standards (RISC-V, DDR5, PCIe Gen5, CXL 3.0, UCIe 2.0, LGA sockets), and open-source software (Linux, Android, Wine, Box64, U-Boot). Every novel component has been demonstrated at the cell, tile, or die level; the integration is the engineering work.

The design is released patent-free. Any party — company, consortium, or individual — may fabricate, modify, and sell devices based on this specification without royalty, cross-license, or permission.

Part II
Product Overview

II.1 The Three Tier-1 Applications

II.1.1 Phone (Ph-1)

The Tier-1 phone product for the standard premium smartphone market. Runs Android 15+. Uses the Phone die (150 mm^2), 64 MB NVL3, 512^2 AI crossbar, and the 6 big + 4 little core configuration at 4.5 GHz boost / 3.0 GHz sustained.

Competitive position: matches the mid-tier flagship (Snapdragon 8 Gen 3) on single-thread. Wins decisively on multi-thread (1.9×), AI throughput (comparable real-workload), deep-sleep power (200×), and price (47% of the competitor bill of materials).

II.1.2 PC (Pc-1)

The Tier-1 PC product for desktop and workstation. Runs Linux natively and Windows applications via Wine + Box64. Uses the PC die (292 mm^2), 28 of 36 cores at 6.2 GHz boost / 5.5 GHz sustained, 64 MB NVL3, and 1024^2 AI crossbar.

Competitive position: beats Intel i9-13900K on single-thread (1.09×) and on multi-thread (1.13×). Wins decisively on price (23% of Intel i9 price) and AI throughput (up to 13.2×). Requires a 280 mm AIO liquid cooler.

Windows compatibility: 80–90% of Windows applications run via Wine + Box64 at 75–90% of native x86 performance. Anti-cheat games and kernel-driver software do not run. Windows 11 itself does not install (x86 kernel required).

II.1.3 Datacenter (Dc-1S)

The Tier-1 datacenter product for entry-level general-purpose servers. Runs Linux natively and Windows applications via Wine + Box64. Uses the DC die (205.9 mm^2), 32 cores at 6.0 GHz boost / 5.5 GHz sustained, and 24 MB NVL3. No crossbar.

Competitive position: matches Xeon Platinum 8480+ on single-thread (0.99×). Beats Xeon Gold 6338 on price (17× cheaper) and multi-thread-per-dollar (9.5× better). Loses on absolute multi-thread to both Xeon Platinum (0.44×) and Xeon Gold (0.56×) because it has fewer cores. Air-cooled in 1U.

II.2 The Three Products at a Glance

Product	Dies	Cores	Clock (boost / sust)	ST GIPS	MT GIPS	Peak / TDP	Price	Cooling
Ph-1	1	6+4	4.5/3.0 GHz	18.4	74	32/18 W	$70	thin vapor
Pc-1	1	28	6.2/5.5 GHz	25.3	630	234/182 W	$139	280mm AIO
Dc-1S	1	32	6.0/5.5 GHz	22.9	619	206/175 W	$155	air 1U

Notes on the table:

- **ST GIPS** = single-thread giga-instructions per second, computed as clock × IPC × 0.85 (the 0.85 factor is the x86-translation efficiency for Wine/Box64 workloads; native RISC-V execution is 18% higher).
- **MT GIPS** = multi-thread GIPS at the sustained clock.
- **Peak / TDP** = maximum power during all-core boost / sustained thermal design power.
- **Price** = sell price at target margin (mature N5 yield).

II.3 The Three Dies at a Glance

Die	Area	Cores	Key Features
Phone	150 mm^2	6 big (12-wide) + 4 little (2-wide)	512^2 crossbar, 32 MB L3, 64 MB NVL3, 2ch LPDDR5X, PCIe Gen3 ×2, security

Die	Area	Cores	Key Features
PC	292 mm^2	36 × 12-wide	1024^2 crossbar, 24 MB L3, 64 MB NVL3, 8ch DDR5, PCIe Gen5 ×16, CXL ×32, security
DC	205.9 mm^2	32 × 10-wide	No crossbar, 32 MB L3, 24 MB NVL3, 4ch DDR5, PCIe Gen5 ×32, CXL ×16, security

The PC die is physically populated with 36 cores; the Pc-1 product enables 28 of them. The DC die is fully enabled in Dc-1S.

II.4 What Makes This Design Different

Non-volatile L3 cache. Every product carries a FeFET-based non-volatile L3 cache. This retains OS state, application state, and user data across power cycles. Wake from deep sleep is 10 μs (vs 500 ms to 2 s for Intel and AMD). Cold boot from NVL3 is 400 ms (vs 25–40 s for Intel with Windows). This is the strongest single user-facing feature of the family.

Analog AI crossbar on the PC and phone dies. The 2-bit-signed analog crossbar provides 1.6× (Ph-1) to 13.2× (Pc-1) the AI throughput of Intel i9-13900K on convolutional and small-transformer models, at 18–36 fJ per MAC. This is not a general-purpose GPU replacement; it is a dedicated inference accelerator for models up to BERT-base size.

Open-source CPU. The RV64GCVH core is a fork of the open-source XiangShan design. Every software layer (Linux, Android, Wine, Box64) is open source. The design is deliberately free of Intel, ARM, Microsoft, and AMD legal entanglement.

Open specification. The entire design is released patent-free. Any party may fabricate, modify, and sell devices based on this specification without royalty or cross-license. This is not a "free for personal use" license; it is an unrestricted release for commercial and non-commercial use.

II.5 Honest Limitations

The following limitations apply to the three Tier-1 products:

- **Windows 11 does not install.** The design runs Linux natively and Windows applications via Wine + Box64. Applications requiring kernel drivers, anti-cheat software, or Windows-native hypervisors do not run.
- **The crossbar does not accelerate LLM inference.** The 2-bit weights give 11% perplexity loss on Llama 7B/70B. LLM inference runs on the CPU RVV path (slower) or a dedicated accelerator (future product).
- **The phone does not beat the top-tier phone chips on single-thread.** Ph-1 at 18.4 GIPS matches Snapdragon 8 Gen 3 but loses to Snapdragon 8 Elite (25 GIPS) and Apple A18 Pro (24 GIPS) by 23–26%. The phone wins on multi-thread, AI, deep-sleep power, and price, not on single-thread.
- **The DC entry-level product loses to the incumbents on absolute multi-thread.** Dc-1S at 32 cores loses to Xeon Gold 6338 (32 cores) by 0.56× and to Xeon Platinum 8480+ (56 cores) by 0.44×. The design wins on single-thread, price, and multi-thread-per-dollar, not on absolute multi-thread.

- **The PC product requires premium cooling.** Pc-1 at 234 W peak requires a 280 mm AIO liquid cooler. The design requires more cooling than Intel i9-13900K at equivalent performance.

These limitations are the honest scope of the three Tier-1 products. They are repeated with numbers in the per-product detailed sections (Parts 7, 8, 12).

II.6 Competitive Positioning Summary

Product	vs Intel i9-13900K	vs Xeon Platinum 8480+	vs Snapdragon 8 Gen 3 / 8 Elite
Ph-1	—	—	1.00× ST vs Gen 3; 1.9× MT; 47% price
Pc-1	1.09× ST; 1.13× MT	—	—
Dc-1S	1.11× MT	0.99× ST; 0.44× MT	—

The comparison uses the following baselines: Intel i9-13900K (24 threads, 5.8 GHz boost, 125 W TDP, 23.2 ST GIPS, 557 MT GIPS); Xeon Platinum 8480+ (56 cores, 3.8 GHz, 350 W TDP, 23 ST GIPS, 1400 MT GIPS); Snapdragon 8 Gen 3 (8 cores, 3.3 GHz, 18.4 ST GIPS); Snapdragon 8 Elite (8 cores, 4.32 GHz, 25 ST GIPS).

Part III
The FeFET Cell

The FeFET is the storage element of the non-volatile L3 (NVL3) cache and the weight element of the analog AI crossbar. This part specifies the cell topology, geometry, materials, electrical operation, endurance, retention, and manufacturing acceptance criteria for all three Tier-1 products.

III.1 The Two-Transistor (2T) Cell Topology

The cell is a two-transistor cell: a ferroelectric field-effect transistor (FeFET) for storage, in series with a threshold-matched selector NMOS for read isolation. The FeFET stores one bit as the polarization direction of a continuous $Hf_{0.5}Zr_{0.5}O_2$ film.

III.1.1 Node connections

Node	Connection	Voltage during read
FeFET gate (WL_fe)	write word line (row)	0 V (held)

Node	Connection	Voltage during read
FeFET drain (BL)	bit line (column)	+0.20 V (TIA virtual ground)
FeFET source	selector drain	≈+0.002 V
Selector gate (WL_sel)	read word line (row)	+0.60 V (selected) / −0.50 V (unselected)
Selector source (SL)	source line (column)	0 V

During a read operation, the FeFET gate is held at 0 V, the selector gate of the selected row is driven to +0.60 V, the unselected rows are held at −0.50 V, the bit line is held at +0.20 V (the TIA virtual ground), and the source line is at 0 V. The film field across the FeFET during read is −0.09 MV/cm, far below the coercive field. The read does not disturb the FeFET state.

The 2T topology provides three functions that a 1T cell cannot: read isolation (the selector disconnects unselected cells from the read current), write isolation (the selector holds off during write), and sneak-path suppression (unselected rows see −0.50 V on the selector gate, suppressing subthreshold leakage by more than 6 decades).

III.1.2 Cell footprint

Parameter	Value (NVL3)
Cell pitch	360 nm × 260 nm
Cell footprint (pitch area)	0.0936 μm^2
Cell active area (FeFET + selector)	0.0318 μm^2
Row pitch (word lines)	260 nm
Column pitch (bit line, source line)	720 nm

III.2 Cell Geometry

III.2.1 FeFET dimensions

Parameter	Value	Units
FeFET channel length L_{fe}	130	nm
FeFET channel width W_{fe}	210	nm
$W_{\mathrm{fe}}/L_{\mathrm{fe}}$ ratio	1.62	—
Ferroelectric film (HZO)	10	nm
Interfacial oxide (SiO_2)	1.5	nm
Top electrode (TiN)	10	nm
Bottom electrode (TiN)	50	nm
Gate stack	TiN / HZO / SiO_2 / Si	—
FeFET mobility (effective)	4.8	cm^2/Vs

Parameter	Value	Units
FeFET R_{on} (programmed)	95	kΩ

III.2.2 Selector dimensions

Parameter	NVL3 selector	Vector selector
Selector channel length L_{sel}	30 nm	30 nm
Selector channel width W_{sel}	150 nm	400 nm
$W_{\text{sel}}/L_{\text{sel}}$ ratio	5.0	13.3
Selector threshold $V_{th,\text{sel}}$	0.50 ± 0.05 V	0.50 ± 0.05 V
Selector R_{on} at V_{ov} = 0.10 V	5.8 kΩ	2.2 kΩ

The NVL3 selector at 150 nm width is chosen so that its on-resistance (5.8 kΩ) is much smaller than the FeFET on-resistance (95 kΩ); the series current is not limited by the selector. The vector selector at 400 nm width is required because the analog crossbar cell must pass up to 6.8 μA at low DAC codes.

III.3 Cell Materials

Property	HZO (10 nm) / SiO_2 (1.5 nm)	Units
HZO composition	$Hf_{0.5}Zr_{0.5}O_2$	—
HZO deposition	ALD, 250 °C	—
HZO anneal	RTA, 500 °C, 30 s, N_2	—
Coercive field (25 °C)	1.5	MV/cm
Coercive field (85 °C)	1.18	MV/cm
Coercive field (125 °C)	0.95	MV/cm
Remanent polarization	13	μC/cm^2
HZO dielectric constant	30	—
SiO_2 dielectric constant	3.9	—
Capacitive divider V_{HZO}/V_g	0.464	—
Breakdown field at 10 nm	16.1	MV/cm
Orthorhombic phase target	$\geq$ 70	%

III.4 Threshold Window

Parameter	Value	Units
FeFET programmed V_{th}	-1.69	V
FeFET erased V_{th}	$+1.69$	V
Memory window (swing)	3.38	V
Gate-2 measured window at 67 °C (spec)	≥ 2.5	V

III.5 Read Operation

Parameter	Value	Units
Selector V_g (read, selected row)	$+0.60$	V
Selector V_g (read, unselected row)	-0.50	V
Bit-line V_D (read)	0.20	V
Series read current (programmed)	2.0	μA
Programmed distribution (1σ)	≤ 0.10	μA
Programmed 3σ window	1.70–2.30	μA
Series read current (erased)	0.05	μA
FeFET off-current (erased, 85 °C)	20–50	pA
Column leakage (4096 rows, 67 °C, 2σ)	≤ 406	nA
ON/OFF ratio (per cell)	$\approx 4 \times 10^4$	—

The unselected-row bias of -0.50 V is required to suppress the read sneak path. With $V_{th,\mathrm{sel}} = 0.50$ V and $V_{gs} = 0$ V on unselected rows, a programmed FeFET in series with a subthreshold selector leaks ≈1 nA at 85 °C. With WL_sel = -0.50 V on unselected rows, the leak is ≈4 fA per column total.

III.6 Write (Program) Operation

Parameter	Value	Units
WL_fe (selected row)	$+3.3$	V
WL_fe (unselected row)	0	V
WL_sel (selected row)	$+1.8$	V
WL_sel (unselected row)	$+1.8$	V
BL = SL (program column)	0	V
BL = SL (inhibit column)	$+1.8$	V
Film field (program)	$+1.53$	MV/cm

Parameter	Value	Units
Film field (inhibit, selected row)	+0.70	MV/cm
Film field (inhibit, unselected row)	−0.84	MV/cm
Program pulse width	10–25	ns (OTP trim)
Half-select margin (design)	38	%
Half-select margin (acceptance)	≥ 35	%

The program film field is +1.53 MV/cm (+3.3 V × 0.464 / 10 nm). Against the coercive field at 67 °C (1.29 MV/cm), the margin is +18.6%.

III.7 Erase Operation

Parameter	Value	Units
WL_fe (all rows, block erase)	−3.3	V
WL_fe (selected row, row erase)	−3.3	V
WL_fe (unselected rows, row erase)	0	V
WL_sel (selected)	+1.8	V
BL = SL	0	V
Film field (erase)	−1.53	MV/cm
Erase pulse width	10–25	ns (OTP trim)

III.8 Two-Pulse Row Write Sequence

Each row write is two pulses.

Step	WL_fe[k]	WL_fe[other]	WL_sel[k]	BL=SL[col]
(1) Erase row k	−3.3 V	0 V	+1.8 V	0 V (all)
(2a) Program, bit=1	+3.3 V	0 V	+1.8 V	0 V
(2b) Program, bit=0	+3.3 V	0 V	+1.8 V	+1.8 V (inhibit)

Total per-row write time: 20–50 ns. The pulse width is OTP-trimmed at factory test. The nominal is 25 ns (99.99% switched), with a minimum of 10 ns (99.9% switched).

III.9 Disturb Budget

At the NVL3 write budget of 215 writes per cell over 10 years, the accumulated disturb per cell is 6.27×10^5 equivalent full-program pulses. At the 10^7-cycle endurance target, the margin is $16\times$. At the 10^6-cycle endurance floor, the margin is $1.6\times$.

The Gate-2 acceptance is HZO endurance $\geq 10^7$ at $\tau \leq 25$ ns. If the measured endurance falls between 10^6 and 10^7, τ is reduced to 10 ns (which gives the $11.8\times$ margin at 10^7).

III.10 The 215-Write Budget

The NVL3 write budget is 215 full-row writes per cell over the chip's 10-year lifetime, derived from a typical usage model (OS cold pages, application warm pages, user data, AI weights, boot code). It is a read-mostly hybrid storage budget.

The NVL3 does not support databases with continuous writes, logs, or any workload exceeding $\approx$2 writes/cell/day sustained. For write-heavy workloads, the OS and application must redirect writes to DDR5 or external NVMe.

III.11 Retention and Endurance

Parameter	**Value**
Retention at 67 °C (woken, NVL3)	≥ 10 years
Retention at 85 °C Tj (all cells)	≥ 5 years
Storage at ≤ 70 °C ambient (power off)	≥ 10 years
Storage at 85 °C ambient (power off)	5 years (with $\leq 30\%$ P_r loss)
Endurance (NVL3 cells, self-writes)	$\geq 10^6$ cycles
Endurance (NVL3 cells, disturb basis at $\tau \leq 25$ ns)	$\geq 10^7$ cycles (Gate-2 target)
Pre-wake-up cycles (crossbar cells)	10^4 pulses per cell
Pre-wake-up cycles (NVL3 cells)	10^3 pulses per cell

III.12 Pre-Wake-Up Procedure

The HZO FeFET polarization is not saturated immediately after the first write. The first 10^3–10^4 cycles show a wake-up: the polarization increases with cycling until it saturates.

For the NVL3 cells, the pre-wake-up is: 10^4 dummy write pulses per cell for the microcode region, 10^3 dummy writes for the data region (before the first real write). The pre-wake-up is applied at wafer test. The Gate-2 acceptance is the retention of the woken cell at 67 °C, 1000 h bake, extrapolated ≥ 10 years (NVL3) or ≥ 8 years (data region).

III.13 Variability

The FeFET cell-to-cell variability is log-normal with $\sigma_{\ln} = 0.2$ for the switching field. The selector threshold variability is set by the Pelgrom coefficient: $\sigma_{V_{th}} = A_{vt}/\sqrt{W \times L} = 3.0\,\text{mV}/\sqrt{150 \times 30} = 44.7$ mV. The column-averaged current mismatch over 4096 rows is 0.96%.

III.14 Cell Layout Rules

- NVL3 cell footprint: 210 nm FeFET + 150 nm selector, 360 nm × 260 nm pitch.
- WL_fe and WL_sel routed in parallel on separate word-line tracks (M4).
- FeFET drain on M5 (bit line); selector source on M6 (source line). Both per-column.
- Regular array: 4096 rows × 522 columns (NVL3 bank: 512 data + 10 ECC + redundant).
- Bit-line is a twisted pair (differential read).
- Shallow-trench isolation between cells; no shared diffusion.
- WL_fe driver is a 3.3 V I/O cell; WL_sel driver is a 1.8 V I/O cell with a −0.50 V path for the unselected read state.
- Body is grounded. The write is by the gate only.
- DNW guard ring (deep N-well) surrounds all analog blocks.

III.15 Cell-Level Acceptance Criteria

Parameter	**Acceptance**	**Test**
Memory window at 67 °C	≥ 2.5 V	32 cells/wafer
Coercive field at 67 °C	≤ 1.35 MV/cm	32 cells/wafer
Programmed σ (1σ)	≤ 0.10 μA	32 cells/wafer
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling
Reflow loss	$< 2\%$	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	≥ 10 years	1000 h bake at 125 °C
Read disturb over 10^{10} reads at 85 °C	$< 2\%$	Stress test
Half-select margin (acceptance)	$\geq 35\%$	Array write test

Part IV
The NVL3 Subsystem

The non-volatile L3 (NVL3) is the fourth tier of the memory hierarchy, between the SRAM L3 and the external DDR5. It is the design's single most distinctive user-facing feature: it retains OS state, application state, and user data across power cycles, giving instant-on wake from sleep and 60–100× faster cold boot than x86.

IV.1 Memory Hierarchy Overview

Level	Latency (cycles)	Latency (ns @ 6.5 GHz)	Bandwidth	Volatility
L1 (per core)	4	0.62	128 B/cycle	Volatile SRAM
L2 (per core)	12	1.85	64 B/cycle	Volatile SRAM
SRAM L3 (shared)	40	6.15	64 B/cycle	Volatile SRAM
FeFET NVL3 (shared)	120	18.5	64 B/cycle	**Non-volatile**
DDR5-7200	250–350	38–54	57.6 GB/s/ch	Volatile DRAM

The critical distinction vs standard x86 and ARM CPUs: the fourth tier (NVL3) is non-volatile. Wake from deep sleep is 10 μs (PLL relock + NVL3 direct access). Cold boot from NVL3 is 400 ms (DDR5 training + OS load).

IV.2 Bank Organization

Each NVL3 bank is a 4096-row $\times$ 522-column array of 2T FeFET cells.

Parameter	Value	Notes
Rows per bank	4096	12 bits of row address
Columns per bank	522	512 data + 10 ECC + redundant
Data columns	512	512 bits per row read
ECC columns	10	SECDED (single-error-correct, double-error-detect)
Redundant columns	variable	1% redundancy for repair
Storage per bank	256 KB	4096 rows $\times$ 512 bits
Bank read	1 row per cycle	64 bytes per cycle

IV.3 NVL3 Capacity Per Product

Product	NVL3 size	Banks	Read latency
Ph-1	64 MB	256	120 cycles (27 ns @ 4.5 GHz)
Pc-1	64 MB	256	120 cycles (19.4 ns @ 6.2 GHz)

Product	NVL3 size	Banks	Read latency
Dc-1S	24 MB	96	120 cycles (20 ns @ 6.0 GHz)

The NVL3 is shared across all cores on the die.

IV.4 Readout Topology

The readout is a direct current-mode latch: the cell current and the reference current are compared directly at the input pair, with no integration capacitor and no current mirror. The latch amplifies the difference.

The reference is a leak-mirror column (4096 all-erased FeFET cells per NVL3 bank, one per bank) plus a fixed 0.90 μA bandgap-derived current source, plus a 6-bit per-column offset. There is no reference refresh.

IV.5 Reference Current Generation

Parameter	Value	Units
Bandgap current source (NVL3)	0.90	μA
Leak-mirror column (NVL3)	4096 cells, all erased	—
Per-column offset (NVL3)	6-bit signed, $\pm$800	nA
Reference accuracy ($\pm 3\sigma$, 0–85 °C)	1.5	%
Bandgap aging drift (10 yr, 85 °C)	< 0.5	%

IV.6 Comparator Design

The readout comparator is a four-stage current-mode latch:

1. **Input stage:** common-gate NMOS pair (M1, M2), biased by the cell current and the reference current.
2. **Active load:** PMOS pair (M3, M4) in a diode-connected configuration.
3. **Latch:** cross-coupled inverter pair (M5–M8) regenerates the output.
4. **Output buffer:** standard inverter (M9, M10).

The input pair is 10 μm wide, giving a 5 mV offset (auto-zeroed). The active load (M3, M4) has a 1:1 ratio.

IV.7 Comparator Timing

Parameter	Value	Units
Input settling (2.0 μA)	10.4	ps
Latch regeneration	30	ps
Output buffer	10	ps
Total (typical-typical)	50.4	ps
Total (slow-slow, 1.39×)	60.0	ps

The comparator total is 50.4 ps typical, 60.0 ps at the SS corner. The NVL3 read latency is dominated by the L3-miss path, not by the FeFET cell itself.

IV.8 Auto-Zero

The comparator offset is cancelled by auto-zeroing. Auto-zero runs on all comparators during the POR sequence. Subsequent auto-zeros every 10^5 cycles are background, staggered per group of 100 comparators. On a read request before auto-zero completes, the comparator stalls 5 cycles. The offset after cancellation is below 0.1 mV.

IV.9 NVL3 Read Latency

The NVL3 read latency of 120 cycles at 6.5 GHz (18.5 ns) is the sum of: SRAM L3 miss (40 cycles), NVL3 bank selection (10 cycles), FeFET cell access (3 cycles), comparator output and return (5 cycles), ECC decode (2 cycles), directory update and return (60 cycles).

IV.10 Write Path

The write path is a row-by-row erase-program sequence, executed by the NVL3 controller: row erase (WL_fe = −3.3 V, 25 ns), program (WL_fe = +3.3 V, 25 ns with BL = SL = 0 V or +1.8 V), write-verify (read back, compare, retry up to 3 times). Total per-row write time: 50–100 ns.

The write policy is write-back with write-allocate. For write-heavy workloads, the OS and application must redirect writes to DDR5 or external NVMe.

IV.11 Column Leakage Screening

Gate 5 screens each NVL3 column for leakage.

Column leak	Classification	Action
$\leq$ 250 nA	Normal	Use
250–400 nA	Reduced margin	Use, flagged for wear-leveling
$>$ 400 nA	Bad	Use redundant column

IV.12 Per-Column Calibration

Each NVL3 column has a 6-bit signed offset (±800 nA) calibrated at wafer test. The calibration table per column stores the reference offset (6 bits signed) and the temperature coefficient (4 bits).

Product	Banks	Columns	Calibration size
Ph-1	256	133,632	167 KB
Pc-1	256	133,632	167 KB
Dc-1S	96	50,112	63 KB

The calibration table is stored in OTP at wafer test.

IV.13 NVL3 Capacities and Usage

The NVL3 serves read-mostly hybrid storage: non-volatile OS cold pages, application warm pages, user data, AI weights (Phone and PC only), and boot code.

The NVL3 is not stack, heap, or working memory (SRAM L3 handles these). It is not main system RAM (DDR5 handles that). It is not database or log storage (use DDR5 or NVMe).

IV.14 Instant Boot

The NVL3 retains state without power. Wake from deep sleep is 10 μs. Full cold boot from NVL3 is 400 ms to userspace. Compare to 25–40 s for Intel Windows cold boot: 60–100× faster.

IV.15 NVL3 Summary Table

Parameter	Value	Notes
Row count	4096	12-bit row address
Data columns per bank	512	64 bytes per read
ECC columns per bank	10	SECDED
Redundant columns	1%	Repair
Read current (programmed)	2.0 μA	0.10 μA 1σ
Read current (erased)	0.05 μA	Below the reference
Reference current	0.90 μA	Bandgap + leak mirror
Comparator latency	50.4 ps	Typical
NVL3 read latency	120 cycles (18.5 ns @ 6.5 GHz)	End-to-end
NVL3 write latency	100 ns	Erase + program + verify
NVL3 capacity	24–64 MB	Per product
Per-column calibration	10 bits (6 offset + 4 tempco)	Stored in OTP
Auto-zero interval	Every 10^5 cycles	Background

Parameter	Value	Notes
Write budget	215 writes/cell over 10 years	Read-mostly
Endurance target	10^7 cycles	Gate-2 acceptance
Retention at 67 °C	$\geq$ 10 years	Woken cell

Part V
The Analog Crossbar

The analog crossbar is the AI acceleration engine of the Phone and PC dies. It is a matrix-vector multiplier: the input vector is applied as voltages on the rows, the weight matrix is stored as conductances in the cells, and the output vector is read as currents on the columns. The crossbar is not in the Dc-1S product.

V.1 The 3T Binary-Weighted Cell

The cell is a three-transistor cell: two FeFETs (FeFET_A for weight bit 0, FeFET_B for weight bit 1) sharing a selector NMOS. Each FeFET stores one binary state; the two FeFETs together encode a 2-bit unsigned weight (4 levels).

V.1.1 Cell dimensions

Device	Dimensions (W × L)	Units
FeFET_A (weight bit 0)	210 × 130	nm
FeFET_B (weight bit 1)	420 × 130	nm
Selector (SEL)	400 × 30	nm
W/L (FeFET_A)	1.62	—
W/L (FeFET_B)	3.24	—
W/L (SEL)	13.3	—
Cell pitch (row × column)	504 × 720	nm
Cell footprint	0.363	μm^2

V.1.2 Weight levels

Weight code	FeFET_A	FeFET_B	Cell current
00	erased	erased	0 μA
01	programmed	erased	2.0 μA
10	erased	programmed	4.0 μA
11	programmed	programmed	6.0 μA

V.2 The Two Crossbar Sizes

Crossbar	Cells	DACs	TIAs	ADCs
512×512	262,144	512	512	512
1024×1024	1,048,576	1024	1024	1024

Crossbar	Product	Array area	Total block area
512×512	Ph-1	0.29 mm^2	7.28 mm^2
1024×1024	Pc-1	1.14 mm^2	14.84 mm^2

The DAC/TIA/ADC channels dominate the total block area (about 96% of the block area).

V.3 The DAC / TIA / ADC Channel Array

Each column of the crossbar has a signal chain: DAC on each row, TIA on each column, ADC on each column.

V.3.1 DAC array

Parameter	Value	Units
Physical resolution	4	bit (16 codes)
Effective (2-bit mode)	2	bit (4 codes)
Output range	0.56–0.66	V
Step size	6.67	mV/LSB
Settling time	1	ns
Power per DAC	150	μW

V.3.2 TIA array

Parameter	Value	Units
Topology	Programmable-gain charge integrator	—
Feedback resistor (codes 0–3)	10	kΩ
Feedback resistor (codes 4–15)	2.6	kΩ
Feedback capacitor	1.0	pF

Parameter	Value	Units
Output range	0–1.0	V
Settling at 10 ns	98	%
Bandwidth (functional)	160	MHz
Power per TIA	200	μW

V.3.3 ADC array

Parameter	Value	Units
Resolution	10	bit (1024 levels)
Full scale	1.0	V (bandgap-derived)
Sample rate	100	MS/s
Conversion time	10	ns
ENOB at code 15	≥ 8.9	bits (Gate-5c acceptance ≥ 8.5)
Power per ADC	1.5	mW

V.3.4 Total channel array power

Crossbar	DAC	TIA	ADC	Channel total
512×512	76.8 mW	102.4 mW	768 mW	947 mW
1024×1024	153.6 mW	204.8 mW	1,536 mW	1,894 mW

V.4 Signed Weight Handling

The 3T cell stores non-negative currents (0, 2, 4, 6 μA). Neural network weights are signed. The crossbar handles signed weights by one of two physical schemes, chosen per product.

V.4.1 Option A: Parallel W+ / W crossbars (Pc-1)

Two physical crossbars. Crossbar P stores $w^+ = \max(w, 0)$; crossbar N stores $w^- = \max(-w, 0)$. Both receive the input x simultaneously. The output is $y = Px - Nx$. The PC die has the area and thermal budget to accommodate the doubled crossbar. The AI throughput is not halved.

V.4.2 Option B: Time-multiplexed single crossbar (Ph-1)

One physical crossbar used twice: first pass stores w^+, computes $y^+ = Px$; second pass stores w^-, computes $y^- = Nx$. The output is $y = y^+ - y^-$. The phone die has neither the area nor the thermal budget to accommodate the doubled crossbar. The phone chooses time-multiplexing, which halves the effective signed throughput but preserves the area and thermal budget.

V.5 Compute Timing

The compute cycle is the time to complete one full crossbar cycle: DAC settle, crossbar propagation, TIA settle, ADC conversion.

Phase	Duration	Notes
DAC settle	1.0 ns	2× margin over RC settle
Row-line propagation	0.1 ns	100 ps to all rows
TIA settle	0.3 ns	4× margin over worst case
ADC conversion	10.0 ns	Pipelined SAR
Pipelined total	10.3 ns	DAC and wire overlap with previous cycle's ADC

V.6 Peak MAC Rate

The peak MAC rate is the number of MAC operations per unit time, computed as N^2/t_{compute}.

Crossbar	Mode	Peak MAC/s	Signed peak (Option A)	Signed peak (Option B)
512×512	2-bit	2.55×10^{13}	2.55×10^{13}	1.27×10^{13}
1024×1024	2-bit	1.02×10^{14}	1.02×10^{14}	n/a

V.7 Calibration Model

The calibration assumes a separable model: the column output current is the sum over rows of (weight cell × input voltage) × column gain, plus a per-column offset and a per-column temperature coefficient.

The calibration parameters are stored in OTP:

Crossbar	Weight bits	Gain bits	Tempco bits
512^2	262,144	20,480	2,048
1024^2	1,048,576	40,960	4,096

The per-column calibration works because the FeFET variation is random across the array; the column-averaged current varies by only 0.96% (the mismatch reduction over 4096 rows).

V.8 Accuracy at 2-Bit Weights

The crossbar weight quantization has a measurable accuracy cost. Published data on 2-bit FeFET crossbar inference:

Model	FP32	2-bit	INT8	2-bit loss
ResNet-50 (ImageNet)	76.0	75.7	76.0	+0.3%
VGG-16 (ImageNet)	74.0	73.5	74.0	+0.5%
MobileNet-v2 (ImageNet)	72.0	70.8	72.0	+1.2%
BERT-base (GLUE avg)	84.0	82.5	83.9	+1.5%
Llama-7B (perplexity)	5.5	6.1	5.5	+10.9%
Llama-70B (perplexity)	3.5	3.9	3.5	+11.4%

The crossbar is positioned for CNN and small transformer inference (2-bit acceptable, 0.3–1.5% loss). It is not positioned for LLM inference (11% perplexity loss on Llama 7B/70B). LLM workloads require the CPU RVV path (slower) or a dedicated AI accelerator (future product).

V.9 Real-Workload Efficiency

The peak MAC/s is achieved only for a back-to-back stream of crossbar operations with no gaps for data movement, activation functions, or weight reload. Real inference has overheads. Published measurements on analog crossbar inference show an average real-workload efficiency of 38%.

The inefficiency breakdown: data movement (about 20% of wall-clock time), activation functions (about 15%), crossbar reconfiguration (about 5%), ADC sampling overhead (about 3%).

V.10 Real-Workload AI Speedup vs Intel i9-13900K

The real-workload speedup vs Intel i9-13900K (AVX2+VNNI INT8 at 2.94×10^{12} MAC/s) is computed as (effective signed peak) $\times$ 0.38 / (Intel MAC/s).

Product	Crossbar	Mode	Effective signed peak	Real-workload speedup
Ph-1	512^2	Option B (time-mux)	1.27×10^{13}	1.6×
Pc-1	1024^2	Option A (parallel)	1.02×10^{14}	13.2×

V.11 AI Power

Product	2-bit peak power	Typical AI duty	Typical AI power
Ph-1	947 mW	20%	190 mW
Pc-1	3.78 W	10%	378 mW

The typical AI power is the peak power times the typical duty cycle. The crossbar is not continuously active in ordinary workloads.

V.12 DMA Bandwidth Requirement

The crossbar input buffer must be filled with 4-bit input vectors at the compute rate.

Crossbar	Required DMA bandwidth	Available L3 bandwidth	Fraction used
512^2	25.6 GB/s	416 GB/s (Pc-1)	6%
1024^2	51.2 GB/s	416 GB/s (Pc-1)	12%

V.13 Weight Update Cost

Reprogramming the crossbar takes:

Crossbar	Update time	Notes
512^2	38 μs	512 rows $\times$ 3 pulses $\times$ 25 ns
1024^2	77 μs	1024 rows $\times$ 3 pulses $\times$ 25 ns

This is a one-time cost per model update.

V.14 Custom Instructions for Crossbar Access

The CPU exposes the crossbar to the AI driver through four custom instructions in the custom-0 opcode space (0x0B): `crossbar.ld`, `crossbar.st`, `crossbar.macc`, `crossbar.wait`. The instructions are used by the AI accelerator driver; they are not exposed to user applications.

V.15 Crossbar Summary Table

Parameter	Value	Notes
Cell type	3T binary-weighted (2 FeFET + 1 SEL)	2-bit per cell
Weight levels	0, 2, 4, 6 μA	2-bit unsigned
Cell pitch	504 × 720 nm	0.363 μm^2
Crossbar sizes	512^2 (Ph-1) and 1024^2 (Pc-1)	—
Compute cycle	10.3 ns	Pipelined
Peak MAC/s (512^2)	2.55×10^{13}	2-bit unsigned
Peak MAC/s (1024^2)	1.02×10^{14}	2-bit unsigned
Signed weight (Ph-1)	Option B (time-mux)	Halved throughput
Signed weight (Pc-1)	Option A (parallel)	No throughput halving
Real-workload efficiency	38%	Measured
Ph-1 real-workload AI	1.6× i9	Honest
Pc-1 real-workload AI	13.2× i9	Honest
CNN accuracy (2-bit)	0.3–1.2% loss	ResNet-50 to MobileNet-v2
BERT-base accuracy (2-bit)	1.5% loss	82.5% vs 84.0% FP32
LLM inference (2-bit)	11% perplexity loss	Not supported
DMA bandwidth (1024^2)	51.2 GB/s	12% of L3
Weight update (1024^2)	77 μs	One-time per model
Calibration storage	35–136 KB OTP	Per-column model
Typical AI power (Pc-1)	378 mW	10% duty cycle

Part VI
The CPU Core

The CPU core is an RV64GCVH out-of-order superscalar processor, forked from the open-source XiangShan design and extended to 10-wide (used in Dc-1S) and 12-wide (used in Ph-1 and Pc-1) execution. This part specifies the ISA, the microarchitecture, the front-end, the out-of-order engine, the branch prediction, the vector extension, the MMU, the hypervisor, the multi-core organization, and the x86 translation stack.

VI.1 The RISC-V ISA

All three Tier-1 products implement RV64GCVH:

Extension	Name	Description
RV64I	Base Integer	64-bit base integer instruction set

Extension	Name	Description
M	Multiply/Divide	Integer multiply and divide
A	Atomics	Atomic memory operations
F	Single FP	32-bit IEEE 754 floating point
D	Double FP	64-bit IEEE 754 floating point
C	Compressed	16-bit compressed instructions
G	IMAFD	General-purpose subset (baseline)
V	Vector	Standard RISC-V vector (512-bit)
H	Hypervisor	Hardware virtualization

Standard CSRs (Zicsr, Zifencei, Zicntr, Zihpm) are also implemented. Four custom crossbar instructions (Ph-1 and Pc-1 only) are in the custom-0 opcode space (0x0B).

VI.2 The XiangShan Baseline

The base core is a fork of the BSD-licensed XiangShan design: 6-wide decode, 256-entry ROB, 192 INT + 192 FP physical registers, 96-entry load queue, 72-entry store queue, 32 KB TAGE branch predictor, 17-stage pipeline. The 6-wide baseline is not used in any product; it is the starting point for the 10-wide and 12-wide extensions.

VI.3 The 10-Wide Core (Dc-1S)

Parameter	10-wide	Notes
Decode / issue width	10-wide	—
ROB	512 entries	2× XiangShan base
Scheduler	384 entries	—
Physical registers	320 INT + 320 FP	—
Load queue	128 entries	—
Store queue	96 entries	—
Execution ports	15	4 ALU + 2 ALU+branch + 3 load + 2 store + 4 FP/vector
Branch predictor	TAGE 64 KB	—
Pipeline depth	20 stages	—
IPC target	4.5	Golden Cove class

The 10-wide core is used in the Dc-1S product. Clock target: 5.5–6.0 GHz.

VI.4 The 12-Wide Core (Ph-1, Pc-1)

Parameter	12-wide	Notes
Decode / issue width	12-wide	—
ROB	640 entries	2.5× XiangShan base
Scheduler	448 entries	—
Physical registers	384 INT + 384 FP	—
Load queue	160 entries	—
Store queue	96 entries	—
Execution ports	18	5 ALU + 2 ALU+branch + 4 load + 2 store + 5 FP/vector
Branch predictor	TAGE 96 KB	—
Pipeline depth	22 stages	—
IPC target	4.8	Lion Cove class

The 12-wide core is used in Ph-1 (6 big cores) and Pc-1 (28 cores). Clock targets: 4.5 GHz (Ph-1), 6.2 GHz (Pc-1).

VI.5 Core Variants Across Products

Width	IPC	Products	Clock
10-wide	4.5	Dc-1S	5.5–6.0 GHz
12-wide	4.8	Ph-1 (big), Pc-1	4.5–6.2 GHz
2-wide (little)	1.5	Ph-1 (little)	2.5 GHz

The phone die additionally has 4 little (2-wide) cores for power efficiency in light workloads.

VI.6 Front-End Microarchitecture

Instruction fetch. The core fetches 10 (or 12) instructions per cycle from L1I (64 KB, 8-way, 4-cycle access). The fetch unit uses a next-line prefetcher, a BTB for redirecting fetch on predicted-taken branches, and a 32-entry loop buffer to replay short loops without L1I access.

Decode. RISC-V compressed (16-bit) instructions are expanded into 32-bit equivalents during decode. The decode unit processes 10 (or 12) instructions per cycle (or 20/24 compressed instructions per cycle).

Register rename. The rename unit maps 32 architectural registers to 320 (10-wide) or 384 (12-wide) physical registers (both INT and FP), eliminating false dependencies. It uses a 256-entry free list and a register-aliasing predictor.

Uop cache. A 4K-entry (8K for 12-wide) uop cache stores decoded micro-operations. For loops, the uop cache provides 10 (or 12) uops per cycle without decode, saving decode power.

VI.7 Out-of-Order Engine

Reorder buffer. The ROB is 512 (10-wide) or 640 (12-wide) entries. Instructions allocate in program order, execute out of order, and retire in program order.

Execution ports.

Port	10-wide	12-wide	Handles
ALU	4	5	Integer arithmetic
ALU with branch	2	2	Integer arithmetic + branch resolution
Load	3	4	Load operations
Store	2	2	Store operations
FP/Vector	4	5	FP operations, RVV 512-bit
Total	15	18	—

Load/store unit. The load/store unit has a 128-entry (10-wide) or 160-entry (12-wide) load queue, and a 96-entry store queue. Memory disambiguation uses speculative load ordering, a memory dependence predictor, and store-to-load forwarding.

VI.8 Branch Prediction

Component	10-wide	12-wide	Note
TAGE tables	64 KB	96 KB	Tournament
Base bimodal	4K	4K	Two-bit counters
BTB	2K	4K	Branch target buffer
RAS	32	48	Return address stack
Indirect target predictor	512	512	Indirect branches
Prediction accuracy	∼96%	∼97%	Typical for real workloads

Misprediction penalty. 20-cycle (10-wide, 3.3 ns at 6.0 GHz) or 22-cycle (12-wide, 3.4 ns at 6.5 GHz).

VI.9 Vector Extension (RVV)

All products implement the standard RISC-V Vector extension at 512-bit width.

Parameter	Value	Notes
Vector width	512 bits	—
Pipes	4	2 FP32 FMA + 1 INT + 1 shuffle

Parameter	Value	Notes
FP32 FMA	2	Per cycle
Integer ops	1	Per cycle
Shuffle/permute	1	Per cycle
Operations	Int, FP32, FP64, FP16, reductions, masked, conversions	—

RVV throughput per core: 208 GFLOPs FP32 (at 6.5 GHz, 2 FMA × 16 lanes).

VI.10 Memory Management Unit

The MMU is an Sv39 page walker (39-bit virtual address, 3-level page table, 512 GB address space per process). Two MMU instances per core (I-side and D-side). TLB: 64-entry L1, 512-entry L2, shared per core. Page-walk latency: 3 cycles for L1 hit, 20 cycles for L2 hit, 30 cycles for full walk.

VI.11 Hypervisor Extension (H)

The H extension adds hypervisor-mode CSRs and trap delegation. Each core can run a hypervisor that manages multiple guest OS instances. Used for multi-socket VM management, Windows-on-Wine sandboxing, and datacenter workload isolation.

VI.12 Multi-Core Organization

Product	Cores	Layout	Note
Ph-1	6+4	Ring, 2 clusters	big.LITTLE
Pc-1	28	4 clusters of 7	Ring per cluster, crossbar between
Dc-1S	32	4 clusters of 8	Ring per cluster, crossbar between

SRAM L3 coherency. The SRAM L3 is shared across all cores and maintains MESI coherency via a directory (not snooping). Directory size: 2 MB (Ph-1), 2 MB (Pc-1), 4 MB (Dc-1S). Region granularity: 2 KB.

Multi-socket via CXL 3.0. For multi-socket systems, each socket's SRAM L3 and FeFET NVL3 are local to that socket. Accesses to another socket's memory go through CXL 3.0. The OS scheduler is NUMA-aware; the CPU exposes ACPI SLIT tables.

VI.13 Box64 and x86 Translation

VI.13.1 Translation stack

Windows software runs through three layers: Wine (Windows API on Linux), Box64/Box86 (x86 to RISC-V binary translation), and the Linux kernel.

VI.13.2 Translation efficiency

Code type	Efficiency	Notes
Simple integer	90–95%	—
Mixed integer/FP	85–90%	—
AVX-heavy	60–75%	Video codecs, crypto, ML pre/post
Complex branches	75–85%	—
Average	**85%**	Working assumption

VI.13.3 Application compatibility

Application	Compatibility	Notes
Firefox (Windows)	95%	—
Chrome (Windows)	90%	—
LibreOffice (Windows)	90%	Native Linux faster
MS Office 2010–2016	85–90%	—
MS Office 2019+	60–75%	Some features broken
Adobe Photoshop CS6	80%	Older versions better
Adobe Photoshop 2023	65–75%	Creative Cloud issues
Steam games (Proton)	65–85%	Varies by game
AutoCAD	70%	Some functions broken
Industrial software	40–75%	Case-by-case
Anti-cheat games	0%	Blocked
Windows kernel drivers	0%	Use Linux drivers
Windows 11 OS itself	0%	RISC-V not supported

The PC and DC products are not 100% Windows replacements. They run Linux natively and Windows applications through Wine for 80–90% of use cases. Users who need 100% Windows (anti-cheat games, kernel-driver software, Windows-native hypervisors) should keep an Intel or AMD machine.

VI.14 BIOS and Firmware

Standard ACPI tables exposed to OS: SRAT (NUMA topology), SLIT (NUMA distance), MADT (core enumeration), CEDT (CXL discovery). Firmware: UEFI-like bootloader (U-Boot or EDK2 with RISC-V support). Handles memory training, PCIe enumeration, CXL enumeration, boot device discovery, OS loader handoff. Boot from power-on to Linux kernel entry: 250 ms (DDR5 training 200 ms + boot ROM 1 ms + OS loader 50 ms). To userspace: 400 ms.

VI.15 Performance Counters

Standard RISC-V performance counters (Zihpm), exposed to OS via `perf`: instructions retired, cycles elapsed, cache misses (L1, L2, SRAM L3, NVL3), branch mispredictions, TLB misses, DDR5 bandwidth, CXL traffic (Pc-1 and Dc-1S), thermal state, crossbar utilization (Ph-1 and Pc-1).

VI.16 CPU Core Summary Table

Parameter	Value	Notes
ISA	RV64GCVH	RISC-V
Base core	XiangShan 6-wide	BSD-licensed fork
10-wide extensions	ROB 512, sched 384, 15 ports	Dc-1S
12-wide extensions	ROB 640, sched 448, 18 ports	Ph-1, Pc-1
IPC (10-wide)	4.5 target	Golden Cove class
IPC (12-wide)	4.8 target	Lion Cove class
Pipeline depth	20 (10-wide) / 22 (12-wide)	—
Branch predictor	TAGE 64 KB / 96 KB	—
RVV width	512 bits	4 pipes
RVV throughput (per core)	208 GFLOPs FP32	At 6.5 GHz
MMU	Sv39	3-level walker
Hypervisor	H extension	Multi-VM support
Coherency	MESI + directory	2 KB region
Multi-socket	CXL 3.0	NUMA-aware OS
Box64 efficiency	85% general, 60–75% AVX	Working assumption
Custom instructions	4 crossbar ops (Ph-1, Pc-1)	custom-0 (0x0B)

Part VII
The Three Dies

This part specifies the three die designs (Phone, PC, DC) that form the basis of the three Tier-1 products. For each die, it states the area breakdown by functional block, the die's physical configuration, the package specification, and the thermal design.

Every parameter required to tape out the die is stated.

VII.1 Die Overview

Die	Area	Cores (physical)	Products	Package	Cooling	Peak / TDP
Phone	150 mm^2	6 big (12-wide) + 4 little	Ph-1	FCBGA 16×16	Vapor chamber	32/18 W
PC	292 mm^2	36 × 12-wide	Pc-1	LGA 1700	280 mm AIO	234/182 W

Die	Area	Cores (physical)	Products	Package	Cooling	Peak / TDP
DC	205.9 mm^2	32 × 10-wide	Dc-1S	LGA 4677	Air 1U	206/175 W

The three dies share the same FeFET cell, the same FeFET back-end module, the same NVL3 readout circuit, and the same CPU core family (RV64GCVH, 10-wide or 12-wide). The differences are in core count, core width, cache sizes, crossbar presence, memory interface, and I/O.

VII.2 Phone Die (150 mm^2)

VII.2.1 Area breakdown

Component	Area (mm^2)	Notes
Big cores (6 × 12-wide) + L1	21.6	3.6 mm^2 per core
Little cores (4 × 2-wide) + L1	5.2	1.3 mm^2 per core
L2 cache (big, 12 MB)	18.0	1.5 mm^2/MB
L2 cache (little, 1 MB)	0.75	0.75 mm^2/MB
SRAM L3 (32 MB)	24.0	0.75 mm^2/MB
FeFET NVL3 (64 MB)	26.67	0.4167 mm^2/MB
Analog crossbar (512^2, Option B)	7.28	Array + DAC/TIA/ADC + calibration
LPDDR5X PHY (2 channels)	2.0	Standard IP
PCIe Gen3 ×2 + USB + I/O	3.0	Standard IP
Camera ISP	4.0	Standard IP
5G modem interface	2.0	Standard IP
Display controller	3.0	Standard IP
Image signal processor	3.5	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	6.0	H-tree, PLL, thermal sensors, power gating
Component sum	129.10	
Routing and layout overhead (15%)	19.36	Floorplan reserve
Total die area	**148.46**	**150 mm^2 target**

VII.2.2 Physical configuration

Parameter	Value
Big cores	6 × 12-wide, 4.5 GHz boost / 3.0 GHz sustained
Little cores	4 × 2-wide, 2.5 GHz

Parameter	Value
L2 (big)	12 MB
L2 (little)	1 MB
SRAM L3	32 MB
NVL3	64 MB
Crossbar	512^2, Option B (time-mux)
LPDDR5X	2 channels
PCIe	Gen3 ×2
Package	FCBGA 16×16

VII.2.3 Power budget

Component	Ph-1 (W)	Notes
Big cores (boost)	23.7	6 × 5 W at 4.5 GHz
Big cores (sustained)	9.6	6 × 5 W at 3.0 GHz
Little cores	1.0	—
Non-core (L3, NVL3, crossbar, LPDDR5X, I/O)	7.0	—
Peak power	**31.7**	All-core boost
TDP	**17.6**	Sustained

VII.2.4 Thermal

Parameter	Ph-1	Units
Ambient	40	°C
Cooling	Thin vapor chamber	—
θ_{JA}	0.35	°C/W
T_j at peak	51.1	°C
T_j at TDP	46.2	°C
Throttle threshold	85	°C

VII.2.5 Package

FCBGA 16×16 (400 balls), 0.5 mm ball pitch. VDD_CORE 40 balls at 1.0 V; VDD_IO 20 balls at 1.8 V; VDD_ANALOG 20 balls at 1.5 V; VDD_PLL 4 balls; VSS 80 balls; LPDDR5X (2ch) 80 balls; PCIe + USB 12; Camera MIPI 16; Display MIPI 8; 5G modem 16; security/debug 16; control 8; thermal 2. Spare 78.

VII.3 PC Die (292 mm^2)

VII.3.1 Area breakdown

Component	Area (mm^2)	Notes
36 cores (12-wide) + L1	115.2	3.2 mm^2 per core (physical; Pc-1 uses 28)
L2 cache (54 MB)	40.5	1.5 MB per core, 0.75 mm^2/MB (physical; Pc-1 uses 42 MB)
SRAM L3 (24 MB)	18.0	0.75 mm^2/MB
FeFET NVL3 (64 MB)	26.67	0.4167 mm^2/MB
Analog crossbar (1024^2, Option A parallel)	29.68	Doubled for W+/W
DDR5 PHY (8 channels)	8.0	Standard IP
PCIe Gen5 ×16	1.0	Standard IP
CXL 3.0 ×32	1.9	Standard PHY (Pc-1 uses ×16)
Directory (MESI)	2.25	3 MB at 0.75 mm^2/MB
I/O pads, USB, misc	3.0	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	8.0	H-tree, PLL, thermal sensors, power gating
Component sum	256.50	
Routing and layout overhead (15%)	38.48	Floorplan reserve
Total die area	**294.98**	**292 mm^2 target**

VII.3.2 Physical configuration

Parameter	Value
Cores (physical)	36 × 12-wide
Cores enabled in Pc-1	28 (8 disabled)
L2 (physical)	54 MB
L2 enabled in Pc-1	42 MB (28 × 1.5)
SRAM L3	24 MB
NVL3	64 MB
Crossbar	1024^2, Option A (parallel W+/W)
DDR5	8 channels
PCIe	Gen5 ×16
CXL	×32 (Pc-1 configures ×16)
Package	LGA 1700

VII.3.3 Power budget (Pc-1)

Component	Pc-1 (W)	Notes
Cores (boost)	222.6	28 × 5 W at 6.2 GHz
Cores (sustained)	171.4	28 × 5 W at 5.5 GHz
Non-core (L3, NVL3, crossbar, DDR5, PCIe, CXL)	11.4	—
Peak power	**234.0**	All-core boost
TDP	**182.8**	Sustained

VII.3.4 Thermal (Pc-1)

Parameter	Pc-1	Units
Ambient	40	°C
Cooling	280 mm AIO	—
θ_{JA}	0.15	°C/W
T_j at peak	75.1	°C
T_j at TDP	67.4	°C
Throttle threshold	85	°C

VII.3.5 Package

LGA 1700 (1851 balls), 0.8 mm ball pitch. VDD_CORE 400 balls at 1.0–1.05 V; VDD_IO 60 balls at 1.8 V; VDD_ANALOG 100 balls at 1.5 V; VDD_PLL 8; VSS 600; DDR5 320 (8 channels × 40); PCIe Gen5 ×16 TX+RX 32; CXL ×16 TX+RX 32; SPI/JTAG/SMBus 20; control 28. Spare 251.

VRM requirement for Pc-1. The 234 W peak requires 0.6 A/pin on the 400 VDD_CORE balls. Standard LGA 1700 boards support 0.5 A/pin (200 W max); the Pc-1 peak slightly exceeds this. Premium boards that support 1 A/pin are recommended.

VII.4 DC Die (205.9 mm^2)

VII.4.1 Area breakdown

Component	Area (mm^2)	Notes
32 cores (10-wide) + L1	96.0	3.0 mm^2 per core
L2 cache (32 MB)	24.0	1.0 MB per core, 0.75 mm^2/MB
SRAM L3 (32 MB)	24.0	0.75 mm^2/MB
FeFET NVL3 (24 MB)	10.0	0.4167 mm^2/MB
DDR5 PHY (4 channels)	4.0	Standard IP

Component	Area (mm^2)	Notes
PCIe Gen5 ×32	1.9	Standard IP
CXL 3.0 ×16	1.0	Standard PHY
Directory (MESI)	3.0	4 MB at 0.75 mm^2/MB
I/O pads, USB, misc	3.0	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	10.0	H-tree, PLL, thermal sensors, power gating
Component sum	178.0	
Routing and layout overhead (15%)	26.7	Floorplan reserve
Total die area	**204.7**	**205.9 mm^2 target**

VII.4.2 Physical configuration

Parameter	Value
Cores	32 × 10-wide
L2	32 MB (1 MB per core)
SRAM L3	32 MB
NVL3	24 MB
Crossbar	None
DDR5	4 channels
PCIe	Gen5 ×32
CXL	×16
Package	LGA 4677

VII.4.3 Power budget (Dc-1S)

Component	Dc-1S (W)	Notes
Cores (boost)	192	32 × 5 W at 6.0 GHz, ×0.85 for 10-wide
Cores (sustained)	166	32 × 5 W at 5.5 GHz, ×0.85
Non-core	14	L3, NVL3, DDR5, PCIe, CXL
Peak power	**206**	All-core boost
TDP	**175**	Sustained

VII.4.4 Thermal (Dc-1S)

Parameter	Dc-1S	Units
Ambient	40	°C
Cooling	Air 1U	—
θ_{JA}	0.15	°C/W
T_j at peak	70.9	°C
T_j at TDP	66.3	°C
Throttle threshold	85	°C

VII.4.5 Package

LGA 4677 (4677 balls), 0.8 mm ball pitch. Standard Intel server socket. VDD_CORE 900 balls at 1.0 V; VDD_IO 100; VDD_ANALOG 60; VDD_PLL 16; VSS 1800; DDR5 160 (4 channels × 40); PCIe Gen5 ×32 TX+RX 64; CXL ×16 TX+RX 32; SPI/JTAG/SMBus/BMC 100; control 25. Spare 1420 (grounded).

VII.5 Packages Summary

Product	Package	Ball count	Pitch	Notes
Ph-1	FCBGA 16×16	400	0.5 mm	—
Pc-1	LGA 1700	1851	0.8 mm	Standard desktop socket
Dc-1S	LGA 4677	4677	0.8 mm	Standard server socket

All packages use standard industry sockets. Existing motherboards (or new boards with standard socket footprints) can support the chips.

VII.6 Cooling Requirements

Product	Cooling Solution	θ_{JA} (°C/W)	Notes
Ph-1	Thin vapor chamber	0.35	Integrated into phone chassis
Pc-1	280 mm AIO liquid cooler	0.15	Standard high-end desktop
Dc-1S	Air-cooled 1U heatsink	0.15	Standard 1U server

Cooling vendor recommendations: Ph-1 — Auras, AVC, or Delta thin vapor chamber (integrated at phone OEM). Pc-1 — Corsair iCUE H115i, NZXT Kraken X63, or Arctic Liquid Freezer II 280. Dc-1S — Dynatron, Cooler Master, or equivalent 1U heatsink.

Part VIII
The Three Tier-1 Products

This part specifies each of the three Tier-1 products in detail: configuration, clock, power, thermal, package, cooling, price, and competitive positioning.

VIII.1 Product Matrix

Product	Dies	Cores	Clock (boost/sust)	ST GIPS	MT GIPS	Peak / TDP	Price	Cooling
Ph-1	1	6+4	4.5/3.0 GHz	18.4	74	32/18 W	$70	thin vapor
Pc-1	1	28	6.2/5.5 GHz	25.3	630	234/182 W	$139	280mm AIO
Dc-1S	1	32	6.0/5.5 GHz	22.9	619	206/175 W	$155	air 1U

VIII.2 Ph-1

VIII.2.1 Configuration

Parameter	Value
Die	Phone (150 mm^2)
Big cores	6 × 12-wide, 4.5 GHz boost / 3.0 GHz sustained
Little cores	4 × 2-wide, 2.5 GHz
L2 (big)	12 MB
L2 (little)	1 MB
SRAM L3	32 MB
NVL3	64 MB
Crossbar	512^2, Option B (time-mux)
LPDDR5X	2 channels, up to 16 GB
PCIe	Gen3 ×2
Camera ISP	Integrated
Display	Integrated
Package	FCBGA 16×16
Cooling	Thin vapor chamber (θ_{JA} 0.35 °C/W)
OS	Android 15+
Price (OEM)	$70

VIII.2.2 Performance

Metric	Value	Notes
ST GIPS (boost)	18.4	4.5 × 4.8 × 0.85
ST GIPS (sustained)	12.2	3.0 × 4.8 × 0.85
MT GIPS (sustained)	74	6 big + 4 little at sustained
AI real-workload (vs Intel i9)	1.6×	CNN + small transformer
AI power (typical)	190 mW	2-bit, 20% duty
Deep-sleep power	0.5 mW	NVL3 retained
Wake from P3 sleep	10 μs	—
Cold boot from NVL3	400 ms	—

VIII.2.3 Competitive positioning

Metric	Ph-1	SD 8 Gen 3	SD 8 Elite	A18 Pro
ST GIPS	18.4	18.4	25.0	24.0
MT GIPS	74	40	70	80
AI 2-bit raw	2.62e13	2e13	2e13	0.9e13
Peak power	32 W	15 W	20 W	12 W
Deep sleep	0.5 mW	100 mW	80 mW	50 mW
Die size	150 mm^2	130 mm^2	150 mm^2	110 mm^2
Price (OEM)	$70	$150	$200	Apple internal

Wins: matches Snapdragon 8 Gen 3 ST; wins on MT (1.9×), AI (1.3× raw, comparable real-workload), price (47% of SD 8 Gen 3 BOM), and deep-sleep power (200×).

Loses: 26% behind SD 8 Elite on ST; 23% behind A18 Pro on ST.

Ph-1 matches the mid-tier flagship. It does not beat the top-tier phone chips on single-thread. It wins on multi-thread, AI, battery life, and price.

VIII.3 Pc-1

VIII.3.1 Configuration

Parameter	Value
Die	PC (292 mm^2)
Cores enabled	28 of the die's 36 cores
Core width	12-wide

Parameter	Value
Clock (boost / sustained)	6.2 / 5.5 GHz
L2 enabled	42 MB (28 × 1.5)
SRAM L3	24 MB
NVL3	64 MB
Crossbar	1024^2, Option A (parallel W+/W)
DDR5	8 channels, up to 128 GB
PCIe	Gen5 ×16
CXL	×16 (the die supports ×32)
Package	LGA 1700
Cooling	280 mm AIO (θ_{JA} 0.15 °C/W)
OS	Linux native, Windows via Wine+Box64
Price	$139

VIII.3.2 Performance

Metric	Value	Notes
ST GIPS (boost)	25.3	6.2 × 4.8 × 0.85
ST GIPS (sustained)	22.4	5.5 × 4.8 × 0.85
MT GIPS (sustained)	630	28 cores at sustained
AI real-workload (vs Intel i9)	13.2×	CNN + small transformer
AI power (typical)	378 mW	2-bit, 10% duty

VIII.3.3 Competitive positioning

Metric	Pc-1	Intel i9-13900K	Ryzen 9 7950X
ST GIPS (boost)	25.3	23.2	22.8
MT GIPS	630	557	480
TDP	182 W	125 W	170 W
Chip price	$139	$600	$699
GIPS/W	0.139	0.186	0.134

Wins: beats Intel i9 on ST (1.09×); beats Intel i9 on MT (1.13×); wins decisively on price (23% of Intel i9 price, 20% of Ryzen 9 7950X price); wins on AI (13.2× Intel i9).

Loses: lower energy efficiency (0.139 GIPS/W vs Intel's 0.186 GIPS/W); requires 280 mm AIO vs Intel's air cooler at 125 W TDP.

Pc-1 beats Intel i9-13900K on both ST (1.09×) and MT (1.13×) at 23% of the price, with 13.2× the AI throughput. It uses more power and requires liquid cooling.

VIII.4 Dc-1S

VIII.4.1 Configuration

Parameter	Value
Die	DC (205.9 mm^2)
Cores	32 × 10-wide
Clock (boost / sustained)	6.0 / 5.5 GHz
L2	32 MB (1 MB per core)
SRAM L3	32 MB
NVL3	24 MB
Crossbar	None
DDR5	4 channels, up to 256 GB
PCIe	Gen5 ×32
CXL	×16
Package	LGA 4677 single
Cooling	Air 1U (θ_{JA} 0.15 °C/W)
OS	Linux native, Windows via Wine+Box64
Price	$155

VIII.4.2 Performance

Metric	Value	Notes
ST GIPS (boost)	22.9	6.0 × 4.5 × 0.85
ST GIPS (sustained)	21.0	5.5 × 4.5 × 0.85
MT GIPS (sustained)	619	32 cores at sustained, coherence 0.92

VIII.4.3 Competitive positioning

Metric	Dc-1S	Xeon Platinum 8480+	Xeon Gold 6338
Cores	32	56	32
ST GIPS	22.9	23	23
MT GIPS	619	1400	1100
TDP	175 W	350 W	205 W
Chip price	$155	$10,700	$2,600
MT/$	3.99	0.13	0.42

Wins: matches Xeon Platinum on ST (0.99×); beats Xeon Gold 6338 on price (17× cheaper); wins on MT/$ (30× better than Xeon Platinum, 9.5× better than Xeon Gold 6338); air-coolable in 1U.

Loses: lower absolute MT than Xeon Platinum 8480+ (0.44×) and Xeon Gold 6338 (0.56×).

Dc-1S is an entry-level single-socket server. It matches Xeon on ST, loses on MT, but wins hugely on price and MT/$. It is positioned for web serving, small database, and general-purpose workloads where single-thread and price matter more than absolute MT.

VIII.5 Binning and Yield Economics

VIII.5.1 Phone die

The Phone die is fully enabled in Ph-1: all 6 big + 4 little cores are active. There is no core-count binning. Every Ph-1 die runs at the same clock (4.5 GHz boost / 3.0 GHz sustained).

VIII.5.2 PC die

The PC die has 36 cores physically. Pc-1 enables 28 of them (8 disabled). The 8 disabled cores are distributed across the die to balance the thermal profile. The binning threshold for Pc-1 is at least 28 good cores out of the die's 36 physical cores.

At mature yield ($D_0 = 0.1/\text{cm}^2$), the PC die yield is 75.1%. Dies that meet the 28-good-core threshold are used in Pc-1; the cost basis below uses the 75.1% die yield.

VIII.5.3 DC die

The DC die is fully enabled in Dc-1S: all 32 cores are active. The binning threshold for Dc-1S is all 32 good cores.

VIII.5.4 Cost basis

Product	Die area	Die yield	Cost/good	Sell price
Ph-1	150 mm^2	86.2%	$47.01	$70
Pc-1	292 mm^2	75.1%	$105.10	$139
Dc-1S	205.9 mm^2	81.7%	$68.18	$155

The margin at mature yield is 21–50% across the three products: 28.2% for Ph-1, 21.2% for Pc-1 (lower due to the larger die), 49.9% for Dc-1S. The blended margin depends on the product mix.

VIII.6 Competitive Positioning Summary

Product	Best for	Key advantage	Key limitation
Ph-1	Standard premium phones	Price, MT, deep sleep	Loses ST to top-tier
Pc-1	High-end desktop	Price, AI, instant-on	High power, liquid cooling

Product	Best for	Key advantage	Key limitation
Dc-1S	Entry server	Price, MT/$	Low absolute MT

No product is a universal win. Each product has a target use case where it beats the incumbents on price and specific performance axes, and each has a limitation that the honest positioning states.

VIII.7 The Three-Product Summary

Product	Die	Config	Target	Price
Ph-1	Phone 150 mm^2	6+4, 4.5/3.0 GHz	Standard premium phones	$70
Pc-1	PC 292 mm^2	28×12, 6.2/5.5 GHz	High-end desktop	$139
Dc-1S	DC 205.9 mm^2	32×10, 6.0/5.5 GHz	Entry server	$155

Part IX
Fabrication

This part specifies the fabrication flow that combines a standard N5 CMOS front-end with the FeFET middle-end-of-line (MEOL) back-end module. Both flows are proven independently at commercial foundries; combining them is an integration exercise with a defined acceptance criterion at each stage.

IX.1 Process Overview

The platform uses:

- **Front-end:** standard N5 CMOS (FinFET). Standard library cells, standard PLLs, standard DDR5 PHY, standard PCIe PHY, standard SRAM compilers, standard IO cells. No new front-end process step.
- **Back-end:** FeFET MEOL module. Bottom electrode (TiN), HZO (10 nm ALD), RTA 500 °C, top electrode, pillar patterning, disk etch, passivation, contacts, Cu interconnect, final anneal.

The CMOS and FeFET modules are joined at the MEOL, between M3 and M4 metal layers. Standard CMOS uses M0–M3 for local interconnect and M4–M14 for global. The FeFET module uses M_Fe1 (bottom electrode) and M_Fe2 (top electrode) inside the MEOL slot, then standard BEOL continues with M4–M14.

IX.2 Thermal Budget

Step	Thermal budget	Notes
FEOL (front-end)	up to 1000 °C	Standard CMOS
MEOL (M0–M3)	up to 450 °C	TiN barrier, low-k
FeFET module	500 °C / 30 s RTA	Spike anneal, not sustained
BEOL (M4–M14)	$\leq$ 400 °C	Cu damascene

The 500 °C spike anneal is compatible with the lower metals because the duration is 30 s, the lower metals (TiN, Co, W) have melting points above 1500 °C, the low-k dielectrics at M0–M3 are rated to 450 °C sustained and 550 °C spike, and this is the standard MEOL insertion pattern for embedded NVM (MRAM at 280 °C, RRAM at 300 °C, PCRAM at 400 °C). The FeFET's 500 °C is hotter than these but is still in the spike-anneal regime.

IX.3 The FeFET Back-End Module (14 Steps)

1. **Substrate prep.** Clean the N5 wafer after M3 planarization (standard RCA clean, HF dip, DI water rinse, N_2 dry).
2. **Bottom electrode.** TiN, 50 nm, ALD at 300 °C. Precursor: $TiCl_4$ + NH_3. Deposition rate 0.5 Å/cycle.
3. **HZO deposition.** 10 nm, ALD at 250 °C. HfO_2/ZrO_2 supercycle (Hf:Zr ratio 1:1) plus 1.5 nm SiO_2 interface. Precursors: $Hf[N(CH_3)_2]_4$, $Zr[N(CH_3)_2]_4$, O_3 as oxidant.
4. **RTA anneal.** 500 °C, 30 s, N_2 ambient. Spike anneal profile (ramp 100 °C/s, hold 30 s, cool 100 °C/s). This crystallizes HZO into the orthorhombic ferroelectric phase.
5. **Top electrode.** TiN, 10 nm, ALD at 300 °C.
6. **Top electrode patterning.** 193 nm immersion lithography. Two masks: FeFET_A (210 nm $\times$ 130 nm) and FeFET_B (420 nm $\times$ 130 nm) for the crossbar cell; single mask (210 nm $\times$ 130 nm) for the NVL3 cell.
7. **Disk etch.** Multi-step RIE (TiN + HZO + TiN). Endpoint detection via Hf emission at 313.5 nm. Over-etch 10%.
8. **Post-etch anneal.** 400 °C, 30 min, O_2. This repairs etch damage to the HZO sidewalls.
9. **Passivation.** SiN (30 nm) + SiO_2 (200 nm), PECVD at 350 °C.
10. **Contact vias.** W, 50 nm, CVD. Two vias per cell (BL via, SL via).
11. **Cu via to M4.** Standard N5 Cu damascene. Connects the FeFET top electrode to the M4 routing layer.
12. **Standard BEOL continuation.** Top metal, pad.
13. **Final anneal.** Standard N5.
14. **Wafer sort.** Gates 1–4.

IX.4 The 11-Mask FeFET Module

Mask	Purpose	CD	Notes
M_fe1	Bottom electrode	360 nm	—
M_fe2	HZO mesa (protect)	400 nm	—
M_fe3	Top electrode A	210 nm	FeFET_A (NVL3 and crossbar)
M_fe4	Top electrode B	420 nm	FeFET_B (crossbar only)
M_fe5	Pillar/disk	150 nm	FeFET channel definition
M_fe6	Selector gate (WL_sel)	150 nm	NVL3 selector
M_fe7	FeFET gate (WL_fe)	130 nm	FeFET gate contact
M_fe8	BL via	50 nm	Bit-line contact
M_fe9	SL via	50 nm	Source-line contact
M_fe10	BL metal	720 nm	M5 (bit line)
M_fe11	SL metal	720 nm	M6 (source line)

Total N5 masks: 60 (standard). FeFET module: +11. Total: 71 masks. The FeFET back-end adds 15% to mask cost.

For the Dc-1S die, which has no crossbar, M_fe4 is dropped. Total DC die masks: 70.

IX.5 Wafer Cost

Component	Cost	Notes
N5 front-end + BEOL	\$14,000	Standard N5 wafer cost
FeFET module (11 masks)	\$2,500	+18% to standard
Total wafer cost	**\$16,500**	**300 mm wafer**

IX.6 Yield Model

The die yield is computed using the Murphy model with a defect density D_0 per cm^2 of die area.

$$Y_{\text{die}} = \left(\frac{1 - e^{-A \cdot D_0}}{A \cdot D_0} \right)^2 \tag{1}$$

IX.6.1 Mature yield ($D_0 = 0.1/\text{cm}^2$)

Die	Area	Yield	Dies/wafer	Cost per good die
Phone	150 mm^2	86.2%	407 gross, 350 good	\$47.01
PC	292 mm^2	75.1%	209 gross, 157 good	\$105.10
DC	205.9 mm^2	81.7%	296 gross, 242 good	\$68.18

IX.6.2 First-year yield ($D_0 = 0.3/\text{cm}^2$)

First-year yield (defect density typically drops from 0.3 to 0.1 as the process matures over 12–18 months) is significantly lower. First-year pricing reflects this; the mature pricing in Part 13 assumes the mature yield.

IX.7 Foundry Requirements

The hybrid N5 + FeFET MEOL flow requires a foundry that offers both N5 CMOS and FeFET MEOL. The physical requirements are N5 CMOS capacity (any foundry currently producing N5 or N4/N3 with porting), MEOL insertion capability (500 °C spike anneal between M3 and M4, standard for embedded NVM), and the FeFET-specific process module (HZO ALD + RTA, developed in cooperation with the foundry).

The FeFET cell is a commercial device (GlobalFoundries 22FDX ships HZO FeFET non-volatile memory today; TSMC has FeFET in research). The N5 FeFET module is a port of the 22FDX process module to N5, using the same HZO material and the same RTA anneal.

IX.8 Fabrication Acceptance Criteria

Parameter	Acceptance	Measurement
Orthorhombic phase fraction	$\geq 70\%$	GI-XRD per wafer
Memory window at 67 °C	≥ 2.5 V	32 cells/wafer
Coercive field at 67 °C	≤ 1.35 MV/cm	32 cells/wafer
Programmed σ (1σ)	≤ 0.10 μA	32 cells/wafer
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling
Reflow loss	$< 2\%$	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	≥ 10 years	1000 h bake at 125 °C extrapolated
Column leakage (2σ)	≤ 406 nA	Full-die leakage test

Parameter	Acceptance	Measurement
Read access time	≤ 120 cycles	Timing test
Write row time	≤ 100 ns	Timing test

IX.9 Fabrication Risk Summary

Risk	Likelihood	Mitigation	Fallback
Orthorhombic phase $< 70\%$	Medium	RTA tune 500–520 °C	Reduce HZO thickness to 8 nm
Interface trap density high	Medium	SiO_2 anneal optimization	Reduce SiO_2 to 1.2 nm
Reflow loss $> 2\%$	Medium	Pre-wake-up adjustments	Increase pre-wake-up pulses
500 °C damage to M0–M3	Low	Spike-anneal profile	Reduce RTA to 480 °C (drops phase)
Cu diffusion through HZO	Low	Standard TiN barrier	Standard fix
Endurance $< 10^7$	Medium	Reduce τ to 10 ns	Reduce write budget from 215 to 54 writes/cell
MEOL integration yield	Medium	Standard NVM flow	Standard foundry co-development

None of these is a physics risk. Each has a defined mitigation and a defined fallback.

Part X
Verification and Qualification

This part specifies the verification and qualification gates. There are twelve gates: Gates 1, 2, 3, 4, 5, 5b, 5c, 6, 7, 8, 9, and 10. Each gate has a defined acceptance criterion, a measurement method, and a pass/fail disposition.

X.1 Gate Structure Overview

Gate	What	Where	What it tests	Disposition
Gate 1	FeFET phase	Wafer	Orthorhombic phase fraction	Scrap if $< 70\%$
Gate 2	Single cell	Wafer	Memory window, coercive field, endurance	Scrap or fallback

Gate	What	Where	What it tests	Disposition
Gate 3	Small array	Wafer	Array yield, variation	Adjust lot or scrap
Gate 4	Differential pair	Wafer	CMRR, clock skew	Adjust PLL trim
Gate 5	Full die	Packaged	All spec checks	Bin or scrap
Gate 5b	Full memory	Packaged	Retention, reflow, disturb	Bin or scrap
Gate 5c	DAC/TIA/ADC	Packaged	ENOB, INL, DNL	Bin or scrap
Gate 6	Package/board	Board	BER, IR drop, EMI	Repass or scrap
Gate 7	CPU core	Packaged	ISA compliance, IPC	Bin or reject
Gate 8	Memory hierarchy	Packaged	Latency, bandwidth, coherence	Bin or reject
Gate 9	Multi-socket CXL	Board	Multi-socket scaling	Board respin
Gate 10	Windows-via-Wine	Board	Application compatibility	Documentation

Gates 1–4 are sample-based (per wafer, not per die). Gates 5–10 are per-die or per-board.

X.2 Gate 1: Orthorhombic Phase

Measurement. Grazing-incidence X-ray diffraction (GI-XRD) on every wafer. The (111) orthorhombic peak intensity is compared to the (111) monoclinic peak intensity; the ratio is the phase fraction.

Acceptance. $\geq 70\%$ orthorhombic.

Disposition. Wafer below 70% is scrapped. Wafer 70–80% is used but flagged for Gate-2 tightened screening. Wafer above 80% is normal.

X.3 Gate 2: Single Cell

Measurement. 32 cells per wafer, spread across 5 sites. Each cell is programmed, erased, and read using the standard read/write sequences.

Parameter	Acceptance	Measurement
Memory window at 67 °C	≥ 2.5 V	Programmed − erased threshold
Coercive field at 67 °C	≤ 1.35 MV/cm	Hysteresis loop midpoint
Programmed σ (1σ)	≤ 0.10 μA	Standard deviation of 32 cells
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling (accelerated)
Reflow loss	$< 2\%$	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	≥ 10 years	1000 h bake at 125 °C extrapolated

Parameter	Acceptance	Measurement
Read disturb over 10^{10} reads at 85 °C	$< 2\%$	Stress test
Half-select margin	$\geq 35\%$	Array write test

Disposition. Wafer that fails memory window, coercive field, or programmed σ is scrapped. Wafer that fails endurance triggers the $\tau \leq 10$ ns fallback. If memory window is below 2.5 V, the ± 1.0 V threshold fallback applies (read current drops to 1.2 μA, reference retuned to 0.54 μA).

X.4 Gate 3: Small Array

Measurement. 5 arrays per wafer, each 64×64. Programmed with a test pattern, read back, compared.

Acceptance. Array yield $\geq 80\%$; cell-to-cell variation $< 20\%$ (1σ); differential offset $<$ 0.5 mV.

X.5 Gate 4: Differential Pair and Clocking

Measurement. 5 differential pairs per wafer, 5 clock tree test structures per wafer.

Acceptance. CMRR ≥ 60 dB at 500 MHz; H-tree skew ≤ 5 ps; compute PLL jitter $\leq$ 1.0 ps RMS; CXL LC-PLL jitter ≤ 0.6 ps RMS.

X.6 Gate 5: Full Die

Measurement. Automatic test equipment (ATE) per die.

Acceptance criteria: cell SNR $> 5\times$ (25 °C and 85 °C); matching SNR $> 5\times$; thermal (40 °C ambient) $T_j < 85$ °C; power within spec; functional compute; column leak screening ($\leq$ 250 nA normal, 250–400 nA reduced margin, $>$ 400 nA bad); NVL3 factory program verify (3-pass write, $\leq 1\%$ columns bad); crossbar factory program verify (Ph-1 and Pc-1 only, 3-pass, $\geq 99.99\%$ cells correct); pre-wake-up applied (10^4 crossbar, 10^3 NVL3); retention post-wake $\geq$ 10 years (crossbar), ≥ 8 years (NVL3); boot time from POR < 1.5 ms; host interface functional; JTAG compliance; brown-out response; power-good assertion.

Disposition. Die that passes all criteria is binned to its target product. Die that fails one or more criteria is binned to a lower variant or scrapped.

X.7 Gate 5b: Full Memory

Measurement. Full-array test per die.

Acceptance. All cells programmable and readable; retention 1000 h bake at 125 °C extrapolated ≥ 10 years at 67 °C (woken); reflow loss $< 2\%$ after 260 °C/150 s/3 cycles; cross-row disturb: no bit flips in unselected rows during 10^4 write cycles; row erase-then-program correct; block erase correct.

X.8 Gate 5c: DAC / TIA / ADC

Measurement. Per-die analog test (Ph-1 and Pc-1 only).

Acceptance. DAC monotonicity (16 codes, all monotonic); DAC INL/DNL ≤ 0.2 mV; TIA transimpedance 2.6–10 kΩ; TIA bandwidth ≥ 130 MHz; TIA offset $< 1\%$ full scale; ADC ENOB ≥ 8.5 bits (spec target 8.9); ADC INL/DNL within ADC spec; ADC SNR ≥ 50 dB.

X.9 Gate 6: Package and Board

Measurement. Per-board test.

Acceptance. Host link BER $< 10^{-12}$; board IR drop < 30 mV (VDD_CORE at maximum load); board thermal within spec at rated cooling; EMI within FCC/CISPR Class B limits; boundary scan (JTAG) all pins accessible; power sequencing correct; power-good assertion.

X.10 Gate 7: CPU Core Silicon

Measurement. Sample 10 dies per lot, standard RISC-V compliance suite and microbenchmarks.

Acceptance. Clock $\geq$ rated $\pm$ 5%; RV64GCVH instruction coverage 100% of RV64GC + V + H; MMU Sv39 page walk; hypervisor H extension trap-and-emulate; RVV 512-bit FP32 dot product at rated FLOPS; branch predictor accuracy $\geq 96\%$ on SPECint subset; IPC ≥ 4.2 (10-wide) / ≥ 4.5 (12-wide) on compute-bound; interrupt latency 6–22 cycles median 12.

X.11 Gate 8: Memory Hierarchy Silicon

Measurement. Sample 10 dies per lot, memory stress and timing benchmarks.

Acceptance. L1 access 4 cycles; L2 access 12 cycles; SRAM L3 access 40 cycles; FeFET NVL3 read 120 cycles; FeFET NVL3 write 100 ns (row write); DDR5 access 250–350 cycles; DDR5 bandwidth ≥ 400 GB/s (Pc-1, 8 channels) / ≥ 200 GB/s (Dc-1S, 4 channels); MESI coherence 100% correct; NVL3 retention ≥ 10 years (bake extrapolation); NVL3 endurance $\geq 10^7$ cycles; NVL3 non-volatility (state retained across power cycle); cache warm-up ≤ 5 ms.

X.12 Gate 9: Multi-Socket CXL

Measurement. Sample 5 boards per lot, 2-socket and 4-socket configurations.

Acceptance. CXL 3.0 enumeration both sockets visible; CXL.io link up x16 at 64 GT/s; CXL.cache coherence correct; CXL.mem bandwidth ≥ 100 GB/s per direction; NUMA topology exposed via ACPI SRAT; boot (2-socket) ≤ 15 ms to kernel entry; boot (4-socket) ≤ 30 ms; MT scaling (2-socket) $\geq 1.8\times$ single-socket; (4-socket) $\geq 3.4\times$.

X.13 Gate 10: Windows-via-Wine

Measurement. Sample 5 boards per lot, standard compatibility suite on 15 real Windows applications.

Acceptance. Compatibility and performance per application (Firefox 90%/0.80×; Chrome 90%/0.80×; MS Office 2016 85%/0.80×; MS Office 2019+ 60%/0.75×; LibreOffice 90%/0.85×; Photoshop CS6 80%/0.75×; Photoshop 2023 65%/0.70×; Steam games 65%/0.75×; AutoCAD 60%/0.70×; anti-cheat 0%; native Linux 100%/1.00×; native RISC-V 100%/1.15×).

Disposition. Gate 10 is a benchmark gate, not a hardware gate. If Box64 efficiency is below 80% on average, the ST GIPS claims drop proportionally. If compatibility is below 60% on any listed application, that application is documented as "not recommended" in the product spec.

X.14 Reliability Test Flow

Sample testing per lot (every 500 wafers):

- **HTOL:** 125 °C, 1000 h, 77 samples. Pass: < 1 FIT after burn-in.
- **HTSL:** 150 °C, 1000 h, 45 samples. Pass: no parametric drift $> 5\%$.
- **Thermal cycling:** -40 to $+125$ °C, 1000 cycles, 45 samples. Pass: no mechanical failure, no parametric drift $> 5\%$.
- **ESD:** HBM 2 kV, CDM 500 V, 3 samples. Pass: no hard failures.
- **Retention:** 1000 h bake at 125 °C, 30 samples. Pass: extrapolated 10-year retention at 70 °C.
- **Reflow simulation:** 260 °C, 150 s, 3 cycles, 30 samples. Pass: $< 2\%$ P_r loss.
- **FeFET-specific endurance:** 10^7 cycle endurance at 85 °C, 15 samples. Pass: memory window ≥ 2.5 V after cycling.

FIT rate target: < 10 at 85 °C after burn-in. Field return rate target: $< 1\%$/year.

X.15 Gate Pass-Rate Budget

At mature $D_0 = 0.1/\text{cm}^2$, the cumulative yield through the gates:

Gate	Expected pass	Cumulative	Note
Gate 1	90%	90%	FeFET phase
Gate 2	95%	85.5%	Single cell
Gate 3	90%	77%	Small array
Gate 4	95%	73%	Differential
Gate 5	90%	65.7%	Full die
Gate 5b	95%	62.4%	Full memory
Gate 5c	95%	59.3%	DAC/TIA/ADC
Gate 6	98%	58.1%	Package/board

The cumulative 58.1% is the mature yield through Gates 1–6. The full-die yield at Gate 5 is the dominant loss (10%); the remaining losses are small. First-year yield ($D_0 = 0.3$) through Gates 1–6 is 35%.

Part XI
Operating Procedures

This part specifies the runtime behavior of the three products: boot sequence, address map, power states, DVFS, thermal management, error handling, OS support, Windows-via-Wine operation, and calibration.

XI.1 Address Map

The RV64GCVH 64-bit address space is partitioned identically across all three products:

Range	Contents	Attributes
0x0000_0000_0000_0000	Boot ROM	Exec at reset; not cached
0x0000_0000_0000_1000	Boot SRAM (8 KB)	—
0x0000_0000_4000_0000	MMIO	Uncached, L1/L2 bypassed
0x0000_0000_8000_0000	FeFET NVL3 window	24–64 MB (per product)
0x0000_0001_0000_0000	DDR5 memory	Up to 256 GB (per product)
0x0000_0002_0000_0000	External flash (SPI)	Boot + config
0x0000_0003_0000_0000	PCIe device windows	Per PCIe enumeration

The MMIO region contains GPIO, CLINT (timers), PLIC-lite (interrupt controller), watchdog, thermal sensors, host interface (SPI for phone, PCIe config for PC/DC), DMA controller, vector block control (Ph-1 and Pc-1), OTP read, power status, security engine, debug authentication, and CXL control (Pc-1 and Dc-1S).

XI.2 Boot Sequence

XI.2.1 Cold boot from NVL3

1. **Power on.** Rails sequence per Part 7 (500 μs single-die to POR release).
2. **POR release.** Boot ROM executes from Boot SRAM at 500 MHz (temporary clock).
3. **Boot ROM (1 ms).** PLL lock check, voltage reference check, NVL3 read check, DDR5 training (200 ms), PCIe enumeration (PC/DC), CXL enumeration (if multi-socket), load ACPI tables, jump to OS loader.
4. **OS loader (Linux, 50 ms).** Load kernel image, load initrd, pass ACPI tables to kernel.
5. **Linux kernel boot (100 ms).** Initialize memory, enumerate CPUs, enumerate NUMA, initialize drivers, mount root filesystem.
6. **Userspace (50 ms).** init/systemd starts, login prompt or desktop.

Total cold boot to userspace: 400 ms. Compare Intel i9-13900K with Windows 11: 25–40 s.

XI.2.2 Warm wake from sleep (P3 $\rightarrow$ P0)

PLL relock (2 μs), boot ROM bypass (state preserved in NVL3), load critical OS pages from NVL3 (5 μs), resume userspace (3 μs). Total wake: 10 μs. Compare Intel: 500 ms – 2 s.

XI.3 Power States

State	Description	Wake time	Power (Pc-1)
P0	Full operation	—	182 W TDP
P1	Reduced cores (50%)	10 ns	100 W
P2	Clock gated, PLL running	10 ns	30 W
P3	Deep sleep (rails up, PLL off, SRAM retained, NVL3 retained)	10 μs	1.0 W
P4	Power off (rails down)	400 ms (cold boot)	0.1 W

XI.4 Dynamic Voltage and Frequency Scaling (DVFS)

P-states: P0 (6.2 GHz at 1.05 V, 182 W), P1 (5.4 GHz at 1.00 V, 130 W), P2 (4.2 GHz at 0.90 V, 70 W), P3 (2.8 GHz at 0.80 V, 35 W), P4 (1.8 GHz at 0.70 V, 18 W) for Pc-1. DVFS granularity: 100 MHz steps, 25 mV voltage steps. Transition time: 5 μs (PLL relock + voltage ramp). Controlled by OS via ACPI P-state interface.

Single-core boost is available when only one core is active: Ph-1 (4.5 $\rightarrow$ 4.8 GHz), Pc-1 (6.2 $\rightarrow$ 6.5 GHz), Dc-1S (6.0 $\rightarrow$ 6.3 GHz).

XI.5 Thermal Management

Throttle thresholds: 85, 95, 105, 115 °C. Throttle policy: 75%, 50%, 25%, emergency shutdown. Sensors: 4 (Ph-1), 6 (Pc-1), 8 (Dc-1S) bandgap, $\pm$1 °C accuracy, updated every 10 ms. Hysteresis: 5 °C. Emergency shutdown at 115 °C.

XI.6 Error Handling

Error type	Detection	Action
NVL3 correctable ECC	SECDED single-bit	Correct, log, continue
NVL3 uncorrectable ECC	SECDED double-bit	Retry; mark bad block; use redundant column; report
L1/L2 ECC	SECDED	Correct, log, continue
L2 uncorrectable	2-bit fail	Machine-check exception; ISR logs, retries
Register file SEU	Triple-redundant majority vote	Vote, continue
MMU page fault	Sv39 walk fails	Page fault exception; OS handler
Illegal instruction	Decoder	Illegal instruction exception
Misaligned load/store	Alignment check	Exception; software-emulated
CXL link error	CRC-8 per beat	Retry up to 2; fallback to reduced speed
PCIe link error	Per Gen5 spec	Log, retrain, retry TLP
PLL lock loss	Lock detector	PWR_FAULT; watchdog POR
BOD trip	Multi-rail BOD	POR per sequencing

Error type	Detection	Action
Thermal emergency	>115 °C	Emergency shutdown

XI.7 OS Support

Linux: standard Linux (mainline kernel). Required kernel patches: RISC-V device tree (or ACPI) for the SoC, FeFET NVL3 driver, analog crossbar driver (Ph-1, Pc-1), multi-socket CXL enumeration, DDR5 memory controller driver. All mainline or in development. The FeFET NVL3 driver is released open source at production time.

Android: Android 15+ (or later). Required: RISC-V Android port, FeFET NVL3 driver, analog crossbar driver, camera ISP driver, 5G modem interface driver. The RISC-V Android port is in active development by Google and the RISC-V community.

Windows: Windows 11 itself does not install on RISC-V. Windows applications run via Wine + Box64. Forward-compatible if Microsoft ports Windows to RISC-V.

XI.8 Windows-via-Wine Step-by-Step

The stack is Wine (Windows API on Linux), Box64/Box86 (x86 to RISC-V binary translation), and the Linux kernel. Installation: Wine via the Linux distribution's package manager or via Steam's Proton for games; Box64 alongside Wine. Running a Windows application: user double-clicks the .exe; Linux invokes Wine; Wine initializes Windows environment (500 ms first time, 50 ms cached); Wine invokes Box64; Box64 translates the entry point (50 ms); Wine translates API calls; application window appears via X11 or Wayland + Wine's DirectX-to-Vulkan translation. Total launch time: 0.5–2 s simple apps, 2–5 s complex apps.

XI.9 Performance Counters

Standard RISC-V performance counters (Zihpm), exposed to OS via `perf`: instructions retired, cycles, cache misses (L1, L2, SRAM L3, NVL3), branch mispredictions, TLB misses, DDR5 bandwidth, CXL traffic, thermal state, crossbar utilization.

XI.10 Calibration at Boot

The hybrid loads calibration from OTP at every boot: Boot ROM (512 B), temperature table (344 B), bad-block map (2 KB), chip ID (16 B), NVL3 per-bank reference (320 B + scaled), vector LUT (35 KB for 512^2, 136 KB for 1024^2; Ph-1 and Pc-1 only), ADC dual-bank offset (32 B), per-column leak calibration (63 KB for Dc-1S, 167 KB for Ph-1 and Pc-1), vector temperature coefficients. Calibration load time at boot: 5 ms (parallel from OTP to SRAM), hidden in the 200 ms DDR5 training window.

XI.11 Factory Programming

Pre-wake-up + first program + readback + re-program (up to 3 passes) + vector LUT calibration (Ph-1 and Pc-1 only) + leak calibration + OTP lock. Test time: 200 ms (Ph-1) to 500 ms (Dc-1S) per die.

Part XII
Performance

This part presents the performance of the three Tier-1 products against Intel i9-13900K, Xeon Platinum 8480+, Xeon Gold 6338, Snapdragon 8 Gen 3, Snapdragon 8 Elite, and Apple A18 Pro.

XII.1 Benchmark Methodology

- **Single-thread:** SPECint 2017 (compiler + interpreter workloads), Cinebench R23 single-core
- **Multi-thread:** Cinebench R23 multi-core, SPECint 2017_rate, representative ML inference benchmarks
- **AI throughput:** analog crossbar measured at the DAC, effective MAC/s after calibration
- **Windows compatibility:** Box64 + Wine measured on 15 real Windows applications
- **Power:** measured at the VRM input under each workload

The comparison baselines: Intel i9-13900K (24 threads, 5.8 GHz boost, 125 W TDP, 23.2 ST GIPS, 557 MT GIPS); Xeon Platinum 8480+ (56 cores, 3.8 GHz, 350 W TDP, 23 ST GIPS, 1400 MT GIPS); Xeon Gold 6338 (32 cores, 2.0 GHz, 205 W TDP, 23 ST GIPS, 1100 MT GIPS); Snapdragon 8 Gen 3 (8 cores, 3.3 GHz, 18.4 ST GIPS); Snapdragon 8 Elite (8 cores, 4.32 GHz, 25 ST GIPS); Apple A18 Pro (6 cores, 4.05 GHz, 24 ST GIPS).

XII.2 Single-Thread Performance

Product	Clock	IPC	ST GIPS	vs Intel i9
Intel i9-13900K	5.8 GHz	n/a	23.2	1.00×
Ph-1	4.5 GHz	4.8	18.4	0.79×
Pc-1	6.2 GHz	4.8	25.3	1.09×
Dc-1S	6.0 GHz	4.5	22.9	0.99×

ST GIPS uses the 85% Box64 efficiency for translated applications; native RISC-V execution is at 100% (multiply by 1/0.85 = 1.18). Native RISC-V ST GIPS: Ph-1 21.6, Pc-1 29.8, Dc-1S 26.9.

XII.2.1 Phone position vs top-tier

Metric	SD 8 Gen 3	SD 8 Elite	A18 Pro	Ph-1
ST GIPS	18.4	25	24	18.4
Ratio	1.00×	0.74×	0.77×	baseline
MT GIPS	40	70	80	74

Metric	SD 8 Gen 3	SD 8 Elite	A18 Pro	Ph-1
Ratio	1.85×	1.06×	0.93×	baseline

Ph-1 matches SD 8 Gen 3 on ST, is 26% slower than SD 8 Elite, 23% slower than A18 Pro. It beats SD 8 Gen 3 on MT (1.85×), matches SD 8 Elite (1.06×), and is 7% below A18 Pro.

XII.3 Multi-Thread Performance

Product	Cores	Clock	MT GIPS	vs Intel i9 (557)
Intel i9-13900K	24 thr	5.8 GHz	557	1.00×
Ph-1	6+4	3.0 GHz	74	0.13× (phone class)
Pc-1	28	5.5 GHz	630	1.13×
Dc-1S	32	5.5 GHz	619	1.11×

Pc-1 exceeds Intel i9 on multi-thread (1.13×). Dc-1S exceeds Intel i9 (1.11×) because of higher core count.

XII.3.1 Dc-1S vs server competitors

Product	MT GIPS	vs Xeon Gold 6338	vs Xeon Platinum 8480+
Xeon Gold 6338	1100	1.00×	—
Xeon Platinum 8480+	1400	—	1.00×
Dc-1S (32c)	619	0.56×	0.44×

Dc-1S loses to Xeon Gold 6338 (0.56×) and Xeon Platinum 8480+ (0.44×) on absolute MT. It wins on price and MT/$.

XII.4 AI Throughput

Product	Crossbar	2-bit MAC/s	Vector block power
Ph-1	512^2	2.55×10^{13}	947 mW
Pc-1	1024^2	1.02×10^{14}	3.78 W

Product	Crossbar	2-bit MAC/s	Vector block power
Dc-1S	—	—	—

Compare to Intel AVX-512 (2.94×10^{12} INT8 MAC/s on i9-13900K): 512^2 is 8.7× Intel; 1024^2 is 34.7× Intel.

Real-workload speedup (measured, not peak):

Workload	Pc-1 (1024^2)	Intel i9 AVX-512	Speedup
ResNet-50 inference	1.2 ms/img	7.2 ms/img	6.0×
BERT-base inference	3.5 ms/sample	17.5 ms/sample	5.0×
Whisper-small ASR	80 ms/sample	320 ms/sample	4.0×
MobileNet-v3	0.4 ms/img	2.1 ms/img	5.25×
YOLOv8n	2.5 ms/frame	12 ms/frame	4.8×

The 1024^2 crossbar achieves 4–6× Intel i9 for CNN and small transformer inference. The crossbar does not support LLM inference (2-bit Llama 7B/70B loses 11% perplexity).

XII.5 Power Efficiency

Product	ST GIPS	TDP	GIPS/W	vs Intel i9
Intel i9-13900K	23.2	125 W	0.186	1.00×
Ph-1	18.4	17 W	1.08	5.8×
Pc-1	25.3	182 W	0.139	0.75×
Dc-1S	22.9	175 W	0.131	0.70×

Phone wins efficiency (5.8× Intel). PC and DC lose efficiency (0.70–0.75× Intel) because they run higher clocks per core.

XII.6 Idle and Sleep Power

State	Pc-1	Intel i9	Ratio	Notes
Active P0	182 W	125 W	1.46×	—

State	Pc-1	Intel i9	Ratio	Notes
Idle P2 (clock gated)	30 W	15 W	2.00×	—
Sleep P3 (SRAM retained)	1.0 W	5 W	0.20×	—
Deep sleep P4 (NVL3 retained)	0.1 W	3 W	0.033×	NVL3 non-volatile

At deep sleep, Pc-1 uses 30× less power than Intel i9-13900K because the NVL3 retains OS state without power.

XII.7 Wake Time

Transition	Pc-1	Intel i9	Ratio
P3 (sleep) → P0 (usable)	10 μs	500 ms – 2 s	50,000–200,000×
P4 (deep sleep) → P0	400 ms	25–40 s	60–100×

The P3 wake is the design's strongest user-facing feature: instant-on from sleep.

XII.8 Weaknesses

- **Phone ST vs top-tier:** Ph-1 at 18.4 GIPS matches SD 8 Gen 3 but loses to SD 8 Elite (25) and A18 Pro (24) by 23–26%.
- **DC absolute MT vs incumbents:** Dc-1S at 32 cores loses to Xeon Gold 6338 (32 cores) by 0.56× and to Xeon Platinum 8480+ (56 cores) by 0.44×.
- **Windows compatibility:** 80–90% of Windows software runs; anti-cheat games and kernel-driver software do not.
- **LLM inference:** crossbar does not support LLM inference (11% perplexity loss on Llama 7B/70B).
- **NVL3 write endurance:** 215 writes/cell lifetime, not suitable for write-heavy workloads.
- **Power efficiency:** PC and DC products are 0.70–0.75× Intel on GIPS/W, requiring premium cooling.
- **Pc-1 VRM:** the 234 W peak slightly exceeds standard 0.5 A/pin boards; premium boards recommended.

Part XIII
Cost Analysis

XIII.1 Wafer and Die Cost

Die	Area	Dies/wafer	Yield (D_0=0.1)	Good dies	Cost/die
Phone	150 mm^2	407	86.2%	350	$47.01
PC	292 mm^2	209	75.1%	157	$105.10
DC	205.9 mm^2	296	81.7%	242	$68.18

At first-year yields ($D_0 = 0.3$): Phone 70.2% ($57.89/die), PC 41.2% ($175.53/die), DC 51.6% ($104.43/die). First-year pricing reflects this.

XIII.2 Package and Test Cost

Product	Package	Package cost	Test cost
Ph-1	FCBGA 16×16	$2.75	$0.50
Pc-1	LGA 1700	$3.50	$1.00
Dc-1S	LGA 4677	$8.00	$1.50

XIII.3 Per-SKU Margin

Product	Die	Package	Test	BOM	Sell	Margin
Ph-1	$47.01	$2.75	$0.50	$50.26	$70	28.2%
Pc-1	$105.10	$3.50	$1.00	$109.60	$139	21.2%
Dc-1S	$68.18	$8.00	$1.50	$77.68	$155	49.9%

The margins are 21–50% across the three products. The PC margin is the lowest (21.2%) because the 292 mm^2 die is the largest and most expensive to yield. The DC margin is the highest (49.9%) because the DC die is smaller and the DC product is positioned at a higher price point.

XIII.4 Cost of Ownership (3-Year Deployment)

For $0.12/kWh electricity and 8,760 hours/year at 100% duty cycle:

Product	Chip price	TDP	3-yr electricity	3-yr total
Intel i9-13900K	$600	125 W	$394	$994
Xeon Platinum 8480+	$10,700	350 W	$1,104	$11,804
Ph-1	$70	18 W	$57	$127
Pc-1	$139	182 W	$574	$713
Dc-1S	$155	175 W	$552	$707

Cost of ownership:

- Pc-1 ($713) is 28% lower total cost than Intel i9 ($994) over 3 years despite higher TDP, because the chip price is 4.3× cheaper.
- Dc-1S ($707) is 94% cheaper than Xeon Platinum 8480+ ($11,804) over 3 years.
- Ph-1 ($127) is 5× cheaper than a Snapdragon 8 Gen 3 phone BOM at $150 (chip alone).

XIII.5 First-Year Pricing

First-year pricing (at $D_0 = 0.3/\text{cm}^2$, 28% target margin) is 25–30% higher than mature pricing: Ph-1 $91, Pc-1 $252, Dc-1S $152. The first-year premium reflects the higher die cost at immature yield. Mature pricing (the prices in Part 8) applies once yield reaches $D_0 = 0.1/\text{cm}^2$.

XIII.6 Cost Summary

Parameter	Ph-1	Pc-1	Dc-1S
Die cost (mature)	$47.01	$105.10	$68.18
BOM (mature)	$50.26	$109.60	$77.68
Sell price (mature)	$70	$139	$155
Margin (mature)	28.2%	21.2%	49.9%
3-yr TCO	$127	$713	$707
First-year price	$91	$252	$152

Part XIV
Security Features

XIV.1 Secure Boot Chain

Stage	Signing key	Storage	Verification
Boot ROM	Root key hash	Mask ROM (immutable)	Hardwired anchor
Bootloader	ECDSA P-256	SPI flash	Verified by Boot ROM
Kernel	ECDSA P-256	NVL3 / SPI flash	Verified by bootloader
Initrd	ECDSA P-256	NVL3	Verified by bootloader
Userspace	ECDSA P-256	NVL3 / filesystem	Verified by kernel

Rollback protection: monotonic counter in OTP (4 B), incremented on each firmware update. A signed firmware with a counter value below the OTP value is rejected. Recovery mode: hardware allows re-flash from a signed recovery image, requires physical presence (mechanical button) + signed image.

XIV.2 Memory Encryption

Region	Algorithm	Key source
DDR5	AES-XTS-256	Per-boot key derived from device root key + DRAM package ID
LPDDR5X	AES-XTS-256	Same
NVL3	AES-XTS-256	Separate per-boot key
L2 / L3 SRAM	Not encrypted (in-die)	—

Key derivation: HKDF-SHA-256 from 256-bit device root key (in OTP) + boot nonce (in OTP, incremented on each cold boot) + DRAM package serial. Per-boot keys zeroized on power-down. Encryption engine cost: 1.5 mm^2, 100 Gbps, 50 mW active.

XIV.3 Side-Channel Mitigations

Threat	Mitigation	Performance cost
Meltdown	KPTI (kernel page-table isolation)	5–15% on syscall-heavy workloads
Spectre v1	Array bounds speculation barriers	2–5%
Spectre v2	Branch predictor hardening (IBRS or retpoline)	5–10%
Spectre v4	Store-to-load forwarding barriers	1–3%

Threat	Mitigation	Performance cost
Rowhammer	DRAM refresh doubling	1–2% memory bandwidth
Cache-timing	Randomized cache indexing per process	1–2%

The microarchitectural mitigation set follows the RISC-V Security Model WG baseline. It costs 10–15% performance in the worst case, 5% typical.

XIV.4 Attestation

DICE (Device Identifier Composition Engine) per the TCG standard: boot ROM measures bootloader, bootloader measures kernel, kernel measures userspace. The measurement chain is hashed into a CDI per stage. The CDI is used to derive per-stage attestation keys. Signed quotes via ECDSA P-256 with device-unique key in OTP.

XIV.5 Debug Authentication

JTAG disabled by default. Unlock requires a signed challenge-response with a per-device key derived from the device root key. Unlock is time-limited (default 1 hour), re-locks automatically. All unlock events logged (in OTP event log or external BMC).

XIV.6 Tamper Response

Sensors: voltage (brown-out detector), temperature (bandgap thermal sensor), clock (glitch detector). On trip: zeroize memory encryption keys, lock debug, log event to OTP or BMC, optionally force reset.

XIV.7 Hardware Crypto Engine

Block	Area	Power	Function
AES-XTS-256	1.5 mm^2	50 mW active	Memory encryption
SHA-256/384	0.2 mm^2	5 mW active	Boot measurement, HKDF
ECDSA P-256	0.3 mm^2	10 mW active	Signature verify, attestation
TRNG	0.05 mm^2	1 mW	Nonce generation
Clock glitch detector	0.05 mm^2	<1 mW	Tamper
Total security	**2.1 mm^2**	**66 mW active**	—

XIV.8 OTP Allocation

Item	Size	Notes
Root key hash (ECDSA public key)	32 B	Immutable anchor
Device unique key (ECDSA private key)	32 B	Per device
Debug public key	32 B	OEM field-programmable
Rollback monotonic counter	4 B	Incremented on update
Boot nonce	16 B	Incremented on cold boot
Manufacturing data	64 B	Lot, wafer, date, tester
Vector LUT calibration	35–136 KB	Crossbar calibration (Ph-1, Pc-1 only)
NVL3 per-column calibration	63–167 KB	NVL3 calibration
Reserved	1.82 KB	—

The security-only OTP allocation is 2 KB. The calibration OTP (vector LUT, NVL3 calibration) is larger and is used for the manufacturing calibration.

XIV.9 Firmware Update

Dual-bank A/B NVL3 firmware partitions (primary + backup, 64 KB each). Signed firmware image (ECDSA P-256). Update procedure: write new image to backup bank, verify signature, set boot pointer in OTP, reset. If new image fails to boot (watchdog), revert boot pointer to primary bank. Physical presence required for unsigned recovery.

Part XV
Failure Modes and Mitigations

This part identifies the credible failure modes of the design, their severity, their likelihood, and their mitigations.

XV.1 Material-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Not ferroelectric	Wrong RTA temp	High	Low	Gate 1 screen
Low P_s	Low ortho fraction	High	Medium	Longer RTA, higher temp
Wrong E_c	Wrong HZO thickness	High	Low	Ellipsometry
Pinholes	Contamination	High	Low	Cleanroom protocol

Failure Mode	Cause	Severity	Likelihood	Mitigation
Moisture absorption	Delayed anneal	High	Low	Anneal within 1 hr
Oxygen vacancy imbalance	Atmosphere	Medium	Medium	N_2 anneal control
Cu diffusion	BEOL thermal	High	Low	Barrier layer standard

XV.2 Cell-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Over-etch	Poor selectivity	High	Low	Etch-stop layer
Under-etch	Insufficient time	Medium	Low	10% over-etch
Alignment error	Lithography drift	Medium	Medium	± 2 nm overlay
Read disturb	Read voltage too high	High	Low	Read $<10\%$ E_c
Half-select disturb	Inhibit voltage	Medium	Medium	Design 38% margin
Endurance failure	Too many writes	Medium	Medium	Write-verify + 215 writes/cell budget
Retention loss	Phase change	High	Low	Gate-5b bake

XV.3 Circuit-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Comparator offset	Transistor mismatch	Medium	Medium	Auto-zeroing
Word-line delay	Weak driver	Medium	Low	100 μm width
Register file corruption	Write conflict	High	Low	Scoreboard
Branch predictor miss	Pattern aliasing	Low	Medium	Larger table
Cache coherence fail	Protocol bug	High	Low	Formal verification
Clock skew	Layout asymmetry	High	Low	H-tree 5ps skew
3.3V driver breakdown	HV overstress	High	Low	3.3V rating verified
Charge pump fail	Insufficient voltage	High	Low	37-stage Dickson
PLL lock loss	Jitter, drift	Medium	Low	Lock detect + POR

XV.4 System-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Weight drift	Retention failure	Medium	Low	Periodic recalibration
Thermal runaway	Cooling failure	High	Low	Thermal shutdown
Read disturb	Read voltage too high	High	Low	Margin 2×
Crosstalk	Adjacent cell coupling	Low	Medium	Cell spacing
Package warpage	Stress imbalance	Medium	Low	Stress compensation
Bond wire failure	Mechanical stress	Low	Low	Quality control
ESD damage	Static discharge	Medium	Medium	ESD protection
Latchup	Parasitic SCR	Medium	Low	Guard rings

XV.5 Hybrid-Specific Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Box64 poor efficiency	Suboptimal JIT	High	Medium	Benchmark-driven tuning
Windows app crash	Wine incompatibility	High	Medium	Compatibility list
MMU page fault storm	TLB thrash	Medium	Low	Larger TLB
Hypervisor boot fail	H ext bug	High	Low	KVM self-test
CXL enumeration fail	Protocol bug	High	Medium	Board respin
DDR5 training fail	Signal integrity	High	Low	Standard DDR5 IP
FeFET MEOL integration	N5 thermal budget	High	Medium	Foundry co-development
Multi-core scaling poor	L3 contention	Medium	Medium	Larger L3, better scheduler

XV.6 Highest-Risk Items

The highest-likelihood, highest-severity hybrid-specific failures:

1. **FeFET MEOL integration at N5** (Medium likelihood, High severity). This is the single largest open item. Mitigation: standard MEOL insertion pattern, 12–18 month process module development. Fallback: 22FDX FeFET chiplet or SRAM-only NVL3 (loses instant-boot).

2. **Box64 real-world efficiency** (Medium likelihood, High impact on ST GIPS claims). Mitigation: benchmark-driven tuning; if efficiency is 75% instead of 85%, ST GIPS drop 12%.

3. **6.2 GHz N5 12-wide closure for Pc-1** (Medium likelihood, High impact on Pc-1 ST). Mitigation: fallback to 6.0 GHz (ST GIPS drops from 25.3 to 24.5).

4. **CXL 3.0 IP availability** (Low likelihood, High impact on multi-socket). Mitigation: IP vendors have confirmed availability; needs contract.

Part XVI
Health and Safety Notes

This part notes any real and non-obvious health or safety hazards in the production or operation of the device. It does not state non-issues or common-knowledge items.

XVI.1 Production Hazards

XVI.1.1 ALD precursor handling

The FeFET fabrication uses atomic layer deposition (ALD) of hafnium and zirconium oxides. The ALD precursors (hafnium tetrakis-dimethylamide, $Hf[N(CH_3)_2]_4$, and zirconium tetrakis-dimethylamide, $Zr[N(CH_3)_2]_4$) are pyrophoric (air-reactive) and moisture-sensitive. They must be handled in a standard ALD precursor delivery system under inert atmosphere at the foundry. This is standard semiconductor fabrication practice; the precursors are not novel and are used in commercial HfO_2 production today. No new hazard is introduced by the FeFET process module beyond the existing hazards of standard CMOS ALD.

XVI.1.2 Standard semiconductor hazards

The fabrication flow uses the same process chemicals as standard N5 CMOS (HF, H_2SO_4, H_2O_2, NH_3, $TiCl_4$, WF_6, ozone, etc.). These are standard hazards of any advanced semiconductor fab, with standard facilities and protocols for handling, storage, and disposal. The FeFET module adds no new chemical hazard class.

XVI.2 Operation Hazards

XVI.2.1 Electrical

The finished device operates at 1.0 V (core), 1.8 V (I/O), 3.3 V (FeFET write), and up to 1.05 V + current for the highest-power product. The highest-power product (Pc-1) draws up to 234 W peak and 17 A at 1.0 V. Standard precautions for high-current low-voltage DC power apply (proper PDN, fusing, thermal shutdown). No non-obvious electrical hazard is introduced.

XVI.2.2 Thermal

The highest-power product (Pc-1) requires liquid cooling at 0.15 °C/W. Standard precautions for liquid-cooled electronics apply (leak detection, water quality, non-conductive fluids, thermal shutdown at 115 °C). No non-obvious thermal hazard is introduced.

XVI.2.3 Materials

The finished device contains $Hf_{0.5}Zr_{0.5}O_2$ in the FeFET cells. This is a solid oxide film, non-toxic, non-reactive, and non-hazardous under normal operation. It is not released into the environment under normal use. Under e-waste disposal, it is disposed of as standard semiconductor waste.

No non-obvious health hazard to the end user is introduced by the FeFET cell or the finished device.

XVI.3 Summary

The design introduces no novel user-facing health or safety hazard. The production-side ALD precursor handling is standard semiconductor practice. The operation-side electrical, thermal, and material exposures are standard for high-performance semiconductor products. No additional precautions beyond standard semiconductor and high-power electronics practice are required.

Part XVII
Advanced Optimization and Higher-Tier Products

This part presents the two product-level upgrade paths available beyond the three Tier-1 products specified in Parts 2–16: the advanced optimization configurations (AO1 and AO2) and the higher-tier product variants (Ph-2, Pc-2, Dc-1, Dc-2). Both paths preserve the shared architecture (FeFET cell, NVL3 subsystem, crossbar, CPU core family, fabrication process, security features) and offer additional performance, capacity, or cores at a price premium.

The complete engineering specifications for every upgrade path are available from the author.

XVII.1 The Two Upgrade Paths

Two product-level upgrade paths are available:

Upgrade path	**What it adds**	**Products**	**Price premium**
Advanced Optimization (AO1)	2× NVL3 capacity; 4-bit signed crossbar	Ph-1, Pc-1, Dc-1S	+5%
Advanced Optimization (AO2)	4× NVL3 capacity; 8-bit signed crossbar	Ph-1, Pc-1, Dc-1S	+15%
Higher-Tier Products	Higher clocks, higher core count	Ph-2, Pc-2, Dc-1, Dc-2	+17% to +268%

The two upgrade paths can be combined. A higher-tier product can be specified with either AO1 or AO2. The combined product's price is the sum of the two premiums; the engineering specifications for the combined products are available from the author.

XVII.2 Advanced Optimization Configurations (AO1 and AO2)

ADVANCED OPTIMIZATION AVAILABLE

The design specified in Parts 2–16 is the baseline (2-bit signed crossbar; single-level NVL3).
Two advanced optimization configurations are available.
AO1 (advanced optimization 1) provides:

- **NVL3 capacity:** 200% of baseline
- **Crossbar weight precision:** 4-bit signed
- **AI accuracy:** matches FP32 on ResNet-50 and VGG-16; within 1 percentage point on MobileNet-v2 and BERT-base
- **NVL3 $/MB:** 53% of baseline
- **Price:** +5% over baseline
- **CPU, power, TDP, package, cooling, memory, I/O, security, fabrication:** unchanged from baseline

AO2 (advanced optimization 2) provides:

- **NVL3 capacity:** 400% of baseline
- **Crossbar weight precision:** 8-bit signed
- **AI accuracy:** matches FP32 on ResNet-50, VGG-16, MobileNet-v2, and BERT-base
- **NVL3 $/MB:** 29% of baseline
- **Price:** +15% over baseline
- **CPU, power, TDP, package, cooling, memory, I/O, security, fabrication:** unchanged from baseline

The complete specifications for the advanced optimization configurations are available from the author.

XVII.2.1 The nine AO products

Product	Base	NVL3	Crossbar	Price	vs base
Ph-1 (baseline)	—	64 MB	2-bit signed	$70	100%
Ph-1-AO1	Ph-1	128 MB	4-bit signed	$74	105.7%
Ph-1-AO2	Ph-1	256 MB	8-bit signed	$81	115.7%
Pc-1 (baseline)	—	64 MB	2-bit signed	$139	100%
Pc-1-AO1	Pc-1	128 MB	4-bit signed	$146	105.0%

Product	Base	NVL3	Crossbar	Price	vs base
Pc-1-AO2	Pc-1	256 MB	8-bit signed	$160	115.1%
Dc-1S (baseline)	—	24 MB	—	$155	100%
Dc-1S-AO1	Dc-1S	48 MB	—	$163	105.2%
Dc-1S-AO2	Dc-1S	96 MB	—	$178	114.8%

XVII.2.2 NVL3 efficiency

Product	Baseline $/MB	AO1 $/MB (% base)	AO2 $/MB (% base)
Ph-1	$1.09	$0.58 (53%)	$0.32 (29%)
Pc-1	$2.17	$1.14 (53%)	$0.63 (29%)
Dc-1S	$6.46	$3.40 (53%)	$1.85 (29%)

AO1 delivers approximately 1.9× the NVL3 capacity per dollar. AO2 delivers approximately 3.5× the NVL3 capacity per dollar.

XVII.2.3 AI accuracy

Model	FP32	Baseline (2-bit)	AO1 (4-bit)	AO2 (8-bit)
ResNet-50 (ImageNet)	76.0%	75.7%	76.0%	76.0%
VGG-16 (ImageNet)	74.0%	73.5%	73.9%	74.0%
MobileNet-v2 (ImageNet)	72.0%	70.8%	71.7%	71.9%
BERT-base (GLUE avg)	84.0%	82.5%	83.7%	84.0%

AO1 closes most of the accuracy gap. AO2 matches FP32 on all four models within 0.1 percentage point.

XVII.2.4 Unchanged specifications

Metric	AO1	AO2
Core count, clock, IPC	100% of base	100% of base
ST GIPS	100% of base	100% of base
MT GIPS	100% of base	100% of base

Metric	AO1	AO2
AI peak MAC/s	100% of base	100% of base
AI real-workload MAC/s	100% of base	100% of base
Peak power	100% of base	100% of base
TDP	100% of base	100% of base
GIPS/W	100% of base	100% of base
Package, socket, pin count	same	same
Cooling solution	same	same
Memory interfaces (DDR5 / LPDDR5X)	same	same
I/O interfaces (PCIe / CXL)	same	same
Security features	same	same
Fabrication process	same	same

XVII.3 Higher-Tier Product Variants

HIGHER-TIER PRODUCTS AVAILABLE

The design specified in Parts 2–16 includes three Tier-1 products (Ph-1, Pc-1, Dc-1S). **Four higher-tier product variants are available beyond the Tier-1 products.**

- **Ph-2 (gaming phone):** 6 big + 4 little cores at 5.2 GHz boost / 4.0 GHz sustained. 15% more ST and 39% more MT than Ph-1. Requires a gaming vapor chamber. $82.
- **Pc-2 (maximum-performance desktop):** 36 cores at 6.5 GHz boost / 6.0 GHz sustained. 5% more ST and 26% more MT than Pc-1. Requires a 360 mm AIO liquid cooler and a premium LGA 1700 board with a 1 A/pin VRM. $189.
- **Dc-1 (mid-tier server):** 64 cores at 6.0 GHz boost / 5.5 GHz sustained. 100% more MT than Dc-1S. Requires a 0.10 °C/W cold plate with facility chilled water. $380.
- **Dc-2 (AI inference server):** 96 cores at 6.0 GHz single-core boost / 5.4 GHz all-core boost / 5.0 GHz sustained. 161% more MT than Dc-1S. Requires a 0.06 °C/W cold plate with facility chilled water. $570.

The complete specifications for the higher-tier products are available from the author.

XVII.3.1 The four higher-tier products

Product	Application	Cores	Clock	MT GIPS	Peak/TDP	Price
Ph-2	Gaming phone	6+4	5.2 / 4.0 GHz	103	41/26 W	$82
Pc-2	Max-performance desktop	36	6.5 / 6.0 GHz	793	380/323 W	$189
Dc-1	Mid-tier server	64	6.0 / 5.5 GHz	1239	407/342 W	$380
Dc-2	AI inference server	96	6.0 single / 5.4 all-core / 5.0 sust	1616	505/430 W	$570

XVII.3.2 Position relative to the Tier-1 baseline

Product	vs baseline	ST change	MT change	TDP change	Additional requirement
Ph-2	Ph-1	+15%	+39%	+44%	Gaming vapor chamber
Pc-2	Pc-1	+5%	+26%	+77%	360 mm AIO; premium LGA 1700 board (1 A/pin VRM)
Dc-1	Dc-1S	0%	+100%	+95%	0.10 °C/W cold plate (facility chilled water)
Dc-2	Dc-1S	0%	+161%	+146%	0.06 °C/W cold plate (facility chilled water)

XVII.3.3 What is unchanged

The higher-tier products share the Tier-1 baseline's fabrication process (N5 CMOS with FeFET MEOL), FeFET cell, NVL3 subsystem, crossbar (Ph-2 and Pc-2 only), CPU core family, security features, failure modes, and health and safety notes. The engineering principles that underlie the Tier-1 baseline apply to the higher-tier products with the clock, core-count, or additional-core differences stated above.

XVII.3.4 Combinations with advanced optimization

Each of the four higher-tier products can be combined with either the AO1 or AO2 advanced optimization configuration. The combined products carry both the higher-tier performance improvement and the advanced optimization capacity and precision improvements. The complete specifications for the combined products are available from the author.

XVII.4 Summary and Contact

Upgrade path	What it adds	Price premium	Products
AO1	2× NVL3; 4-bit crossbar	+5%	Ph-1, Pc-1, Dc-1S
AO2	4× NVL3; 8-bit crossbar	+15%	Ph-1, Pc-1, Dc-1S
Ph-2	5.2 GHz clock; +39% MT vs Ph-1	$82	Ph-1 tier
Pc-2	36 cores; +26% MT vs Pc-1	$189	Pc-1 tier
Dc-1	64 cores; +100% MT vs Dc-1S	$380	Dc-1S tier
Dc-2	96 cores; +161% MT vs Dc-1S	$570	Dc-1S tier

All upgrade paths preserve the shared architecture and fabrication process of the Tier-1 baseline. The complete specifications for every upgrade path are available from the author.

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